

## Hermite Processes

Limits of normalized sums of functions of Gaussian time series with long-range dependence include fractional Brownian motion, the Rosenblatt process or, more generally, the Hermite processes. We present the limit results in the next Chapter 5. We focus in the present chapter on the limit processes themselves. These limits are called *Hermite processes* because they are related to Hermite polynomials. A convenient way to represent Hermite processes is to use multiple stochastic integrals, which are defined in Appendix B.2.

We introduce Hermite polynomials in Section 4.1 and indicate how they are related to multiple stochastic integrals. We provide in Section 4.2 the integral representation of the Hermite processes in the time domain, in the spectral domain and on an interval.

Key tools are moments, cumulants and diagram formulae, which are given in Sections 4.3 and 4.4. We focus on general multiple stochastic integrals of order two in Section 4.5. Then, in Section 4.6, we treat an important example, namely the Rosenblatt process, and consider its marginal cumulative distribution function in Section 4.7 with tables in Section 4.8. Finally, in Section 4.9, we consider generalized Hermite processes.

### 4.1 Hermite Polynomials and Multiple Stochastic Integrals

We first define Hermite<sup>1</sup> polynomials.

**Definition 4.1.1** The *Hermite polynomial of order 0* is  $H_0(x) = 1$ . The *Hermite polynomial of order  $n$* ,  $n = 1, 2, \dots$ , is defined as

$$H_n(x) = (-1)^n e^{x^2/2} \frac{d^n}{dx^n} e^{-x^2/2}, \quad x \in \mathbb{R}. \quad (4.1.1)$$

One thus has

$$\begin{aligned} H_0(x) &= 1, & H_1(x) &= x, & H_2(x) &= x^2 - 1, & H_3(x) &= x^3 - 3x, \\ H_4(x) &= x^4 - 6x^2 + 3, & H_5(x) &= x^5 - 10x^3 + 15x, & H_6(x) &= x^6 - 15x^4 + 45x^2 - 15, \end{aligned}$$

<sup>1</sup> Pronounced [er MEET].

and so on.<sup>2</sup> Hermite polynomials appear in the expansion

$$F(x, t) := e^{tx - \frac{t^2}{2}} = e^{x^2/2} e^{-\frac{(x-t)^2}{2}} = e^{x^2/2} \sum_{n=0}^{\infty} \frac{t^n}{n!} \left( \frac{d^n}{dt^n} e^{-\frac{(x-t)^2}{2}} \Big|_{t=0} \right) = \sum_{n=0}^{\infty} \frac{t^n H_n(x)}{n!}, \quad (4.1.2)$$

where we used the Taylor's expansion of  $e^{-\frac{(x-t)^2}{2}}$  in  $t$ , and

$$\frac{d}{dt} e^{-\frac{(x-t)^2}{2}} = \left( \frac{d}{dy} e^{-\frac{y^2}{2}} \Big|_{y=x-t} \right) \left( \frac{d}{dt} (x-t) \right) = (-1) \frac{d}{dt} e^{-\frac{y^2}{2}} \Big|_{y=x-t}$$

and, more generally,

$$\frac{d^n}{dt^n} e^{-\frac{(x-t)^2}{2}} = (-1)^n \frac{d^n}{dy^n} e^{-\frac{y^2}{2}} \Big|_{y=x-t}$$

and thus

$$\frac{d^n}{dt^n} e^{-\frac{(x-t)^2}{2}} \Big|_{t=0} = (-1)^n \frac{d^n}{dy^n} e^{-\frac{y^2}{2}} \Big|_{y=x} = (-1)^n \frac{d^n}{dx^n} e^{-\frac{x^2}{2}} = e^{-x^2/2} H_n(x),$$

yielding (4.1.2). In particular,

$$\frac{\partial^n}{\partial t^n} F(x, t) \Big|_{t=0} = H_n(x). \quad (4.1.3)$$

We now relate Hermite polynomials, Gaussian variables and multiple stochastic integrals. As in Appendix B.2, we denote the multiple integral with respect to a real-valued Gaussian measure in (B.2.1) as  $I_k(f)$ , for  $f \in L^2(E^k, m^k)$ , and the multiple integral with respect to an Hermitian Gaussian measure in (B.2.14) as  $\widehat{I}_k(g)$ , for Hermitian  $g \in L^2(\mathbb{R}^k, m^k)$ . The tensor product  $\otimes$  is also considered in Appendix B.2.

**Proposition 4.1.2** (a) (Time domain) Let  $h \in L^2(E, m)$  with  $\|h\|_{L^2(E, m)} = 1$ . Then, for  $k \geq 1$ ,

$$H_k(I_1(h)) = I_k(h^{\otimes k}). \quad (4.1.4)$$

(b) (Spectral domain) Let  $g \in L^2(\mathbb{R}, m)$  be Hermitian with  $\|g\|_{L^2(\mathbb{R}, m)} = 1$  and  $m$  be symmetric. Then, for  $k \geq 1$ ,

$$H_k(\widehat{I}_1(g)) = \widehat{I}_k(g^{\otimes k}). \quad (4.1.5)$$

**Example 4.1.3** (Hermite polynomial and multiple integral of order three) Taking  $k = 3$  in (4.1.4) and using  $I_1(h) = \int_E h(u) W(du)$ , for example, one has

$$\begin{aligned} H_3\left(\int_E h(u) W(du)\right) &= \left(\int_E h(u) W(du)\right)^3 - 3 \int_E h(u) W(du) \\ &= \int_{E^3}' h(u_1) h(u_2) h(u_3) W(du_1) W(du_2) W(du_3). \end{aligned}$$

<sup>2</sup> Another, different definition of Hermite polynomials is to set  $\widetilde{H}_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}$ . The connection to the Hermite polynomials in Definition 4.1.1 is  $\widetilde{H}_n(x) = 2^{n/2} H_n(\sqrt{2}x)$ . Yet another way to define Hermite polynomials is to set them equal to  $H_n(x)/n!$  or  $H_n(x)/\sqrt{n!}$ . It is important therefore to always check which definition is used.

**Remark 4.1.4** If, for example,  $\|h\|_{L^2(E,m)} \neq 1$ , Proposition 4.1.2, (a), can be applied to  $h/\|h\|_{L^2(E,m)}$  to obtain

$$\|h\|_{L^2(E,m)}^k H_k\left(\frac{I_1(h)}{\|h\|_{L^2(E,m)}}\right) = I_k(h^{\otimes k}). \quad (4.1.6)$$

Also, note that in the case of  $E = \mathbb{R}$  and  $m$  being the Lebesgue measure, a Gaussian variable  $I_1(h) = \int_{\mathbb{R}} h(u)B(du)$  in part (a) of Proposition 4.1.2 can be represented as  $\widehat{I}_1(\widehat{g}) = \int_{\mathbb{R}} \widehat{g}(x)\widehat{B}(dx)$  in part (b) with  $\widehat{g}(x) = \widehat{h}(x)/\sqrt{2\pi}$ , where  $\widehat{h}(x) = \int_{\mathbb{R}} e^{ixu}h(u)du$  stands for the Fourier transform of  $h$  and an Hermitian Gaussian measure  $\widehat{B}(dx)$  has the Lebesgue control measure. This follows from (B.2.16).

*Proof* Consider part (a). The proof is by induction in  $k$ . The relation (4.1.4) is trivial for  $k = 1$  since  $H_1(x) = x$ . Suppose the relation holds for  $k$  and consider it for  $k + 1$ . By using the property

$$H_{k+1}(x) = xH_k(x) - kH_{k-1}(x), \quad k \geq 1, \quad (4.1.7)$$

of Hermite polynomials (Exercise 4.1) and the induction assumption, we have that

$$H_{k+1}(I_1(h)) = I_1(h)H_k(I_1(h)) - kH_{k-1}(I_1(h)) = I_1(h)I_k(h^{\otimes k}) - kI_{k-1}(h^{\otimes(k-1)}).$$

The last expression equals  $I_{k+1}(h^{\otimes(k+1)})$  by using the formula (B.2.20) with  $f = h^{\otimes k}$  and  $g = h$ , and the fact that

$$h^{\otimes k} \otimes_1 h = h^{\otimes(k-1)} \int_E h(u)h(u)m(du) = h^{\otimes(k-1)},$$

since  $\|h\|_{L^2(E,m)} = 1$ . The proof of part (b) is analogous by using the formula (B.2.23).  $\square$

**Example 4.1.5** (Connection to Itô's stochastic calculus) Take  $E = \mathbb{R}$ ,  $m(du) = du$  and  $h_t(u) = 1_{[0,t)}(u)/\sqrt{t}$ ,  $t > 0$ , in Proposition 4.1.2, (a), so that

$$I_1(h_t) = \int_{\mathbb{R}} h_t(u)B(du) = \int_{\mathbb{R}} \frac{1_{[0,t)}(u)}{\sqrt{t}}B(du) = \frac{B(t)}{\sqrt{t}},$$

where  $\{B(t)\}$  is a standard Brownian motion. Note, for example, that since  $H_2(x) = x^2 - 1$ ,

$$H_2(I_1(h_t)) = H_2\left(\int_{\mathbb{R}} \frac{1_{[0,t)}(u)}{\sqrt{t}}B(du)\right) = \frac{B(t)^2}{t} - 1.$$

On the other hand, by using the relation (B.2.24) of multiple integrals to Itô's stochastic integrals and Itô's formula, one also has

$$\begin{aligned} I_2(h_t^{\otimes 2}) &= \int_{\mathbb{R}^2}' \frac{1_{[0,t)}(u_1)}{\sqrt{t}} \frac{1_{[0,t)}(u_2)}{\sqrt{t}} B(du_1)B(du_2) \\ &= \frac{2}{t} \int_0^t \left\{ \int_0^{u_2} dB(u_1) \right\} dB(u_2) = \frac{2}{t} \int_0^t B(u_2)dB(u_2) \\ &= \frac{1}{t} (B(t)^2 - t) = \frac{B(t)^2}{t} - 1. \end{aligned}$$

Thus, the relation (4.1.4) indeed holds for  $k = 2$  and  $h_t(u) = 1_{[0,t)}(u)/\sqrt{t}$ .

## 4.2 Integral Representations of Hermite Processes

In this section, we define Hermite processes, provide their various representations and consider their basic properties.

### 4.2.1 Integral Representation in the Time Domain

The following is a definition of Hermite processes in time domain.

**Definition 4.2.1** Let  $k \geq 1$  be an integer and

$$H \in \left(\frac{1}{2}, 1\right). \quad (4.2.1)$$

Set

$$H_0 = 1 - \frac{1-H}{k} \in \left(1 - \frac{1}{2k}, 1\right), \quad (4.2.2)$$

so that  $H = 1 - k(1 - H_0)$ . The Hermite process  $\{Z_H^{(k)}(t)\}_{t \in \mathbb{R}}$  of order  $k$  is defined as

$$Z_H^{(k)}(t) = A_{k,H} \int_{\mathbb{R}^k}' \left\{ \int_0^t \prod_{j=1}^k (s - u_j)_+^{-\left(\frac{1}{2} + \frac{1-H}{k}\right)} ds \right\} B(du_1) \dots B(du_k) \quad (4.2.3)$$

$$= a_{k,H_0} \int_{\mathbb{R}^k}' \left\{ \int_0^t \prod_{j=1}^k (s - u_j)_+^{H_0 - \frac{3}{2}} ds \right\} B(du_1) \dots B(du_k), \quad t \in \mathbb{R}, \quad (4.2.4)$$

where  $B(du)$  is a Gaussian random measure on  $\mathbb{R}$  with the Lebesgue control measure  $du$ ,  $u_+ = \max\{u, 0\}$  and  $A_{k,H}, a_{k,H_0}$  are normalizing constants. The integral  $\int_0^t$  is interpreted as  $-\int_t^0$  when  $t < 0$ . The Hermite process  $\{Z_H^{(k)}(t)\}_{t \in \mathbb{R}}$  is called *standard* if  $\mathbb{E}(Z_H^{(k)}(1))^2 = 1$ .

The parameter  $H_0$  is the one used in the FBM kernel and the parameter  $H$  is the self-similarity parameter of the Hermite process, as noted in Proposition 4.2.2 below. Note that the Hermite process of order one is FBM (see Section 2.6.1) and in this case  $H = H_0$ . The Hermite process of order two is the Rosenblatt process (see Section 2.6.3). Note also that the Hermite process of order  $k$  is defined through the kernel function

$$f_t(u_1, \dots, u_k) = \int_0^t \prod_{j=1}^k (s - u_j)_+^{p-1} ds, \quad (4.2.5)$$

where, in view of (4.2.2),<sup>3</sup>

$$p = H_0 - \frac{1}{2} \in \left(\frac{1}{2} - \frac{1}{2k}, \frac{1}{2}\right). \quad (4.2.6)$$

The following result provides basic properties of Hermite processes.

<sup>3</sup> The parametrization  $p$  is well-suited for manipulating Fourier transforms.

**Proposition 4.2.2** *Hermite processes are well-defined, have stationary increments and are  $H$ -self-similar. The Hermite process of order  $k$  is standard when*

$$A_{k,H} = \left( \frac{H(2H-1)}{k!(B(\frac{1}{2} - \frac{1-H}{k}, \frac{2-2H}{k}))^k} \right)^{1/2} \quad (4.2.7)$$

or

$$a_{k,H_0} = \left( \frac{(1-k(1-H_0))(1-2k(1-H_0))}{k!(B(H_0 - \frac{1}{2}, 2-2H_0))^k} \right)^{1/2}, \quad (4.2.8)$$

where  $B(a, b)$  denotes the beta function (2.2.12).

*Proof* The Hermite process of order  $k$  is well-defined as long as  $\|f_t\|_{L^2(\mathbb{R}^k)} < \infty$ , in which case  $\mathbb{E}|Z_H^{(k)}(t)|^2 = k!\|f_t\|_{L^2(\mathbb{R}^k)}^2$ . Since, for example, for  $t > 0$ ,

$$f_t(tu_1, \dots, tu_k) = t^{(p-1)k+1} f_1(u_1, \dots, u_k) = t^{H-\frac{k}{2}} f_1(u_1, \dots, u_k), \quad (4.2.9)$$

it is enough to show that  $\|f_1\|_{L^2(\mathbb{R}^k)} < \infty$ . In fact,  $\|f_1\|_{L^2(\mathbb{R}^k)}$  can be calculated easily. Arguing as in the proof of Proposition 2.6.15, observe that

$$\begin{aligned} \|f_1\|_2^2 &= \int_{\mathbb{R}} du_1 \dots \int_{\mathbb{R}} du_k \left( \int_0^1 \prod_{j=1}^k (s-u_j)_+^{p-1} ds \right)^2 \\ &= \int_0^1 ds_1 \int_0^1 ds_2 \left( \int_{\mathbb{R}} (s_1-u)_+^{p-1} (s_2-u)_+^{p-1} du \right)^k \\ &= 2 \int_0^1 ds_1 \int_{s_1}^1 ds_2 \left( \int_0^\infty z^{p-1} (s_2-s_1+z)^{p-1} dz \right)^k \\ &= 2 \left( \int_0^\infty v^{p-1} (1+v)^{p-1} dv \right)^k \int_0^1 ds_1 \int_{s_1}^1 ds_2 (s_2-s_1)^{(2p-1)k} \\ &= 2(B(p, 1-2p))^k \int_0^1 ds_1 \int_{s_1}^1 ds_2 (s_2-s_1)^{(2p-1)k} \\ &= \frac{2(B(p, 1-2p))^k}{((2p-1)k+1)((2p-1)k+2)} < \infty, \end{aligned}$$

by using (4.2.6). This also yields the normalizing constants (4.2.7)–(4.2.8) by using (4.2.6) and (4.2.2). The  $H$ -self-similarity follows by using the scaling property (4.2.9) of the kernel function, and stationarity of the increments can be proved by using the fact that

$$f_{t+h}(u_1, \dots, u_k) - f_h(u_1, \dots, u_k) = f_t(u_1 - h, \dots, u_k - h). \quad \square$$

### 4.2.2 Integral Representation in the Spectral Domain

The following result extends the result on the Fourier transform in Proposition 2.6.15 to the kernel function (4.2.5).

**Proposition 4.2.3** Let  $f_t(u_1, \dots, u_k)$ ,  $t, u_1, \dots, u_k \in \mathbb{R}$ , be the function defined in (4.2.5)–(4.2.6). Then,

$$\widehat{f}_t(x_1, \dots, x_k) = \Gamma(p)^k \frac{e^{it(x_1 + \dots + x_k)} - 1}{i(x_1 + \dots + x_k)} \prod_{j=1}^k |x_j|^{-p} e^{-i \operatorname{sign}(x_j) \frac{p\pi}{2}}. \quad (4.2.10)$$

*Proof* The result can be proved in the same way as the result on the Fourier transform in Proposition 2.6.15.  $\square$

An immediate corollary of Proposition 4.2.3 is the spectral representation of Hermite processes. We write the representation using only the parametrization  $H_0$ .

**Proposition 4.2.4** The Hermite process of order  $k$  can be represented as

$$\{Z_H^{(k)}(t)\}_{t \in \mathbb{R}} \stackrel{d}{=} \left\{ b_{k, H_0} \int_{\mathbb{R}^k} \frac{e^{it(x_1 + \dots + x_k)} - 1}{i(x_1 + \dots + x_k)} \prod_{j=1}^k |x_j|^{\frac{1}{2} - H_0} \widehat{B}(dx_1) \dots \widehat{B}(dx_k) \right\}_{t \in \mathbb{R}}, \quad (4.2.11)$$

where  $\widehat{B}(dx)$  is an Hermitian Gaussian random measure on  $\mathbb{R}$  with control measure  $dx$ . The normalizing constant  $b_{k, H_0}$  satisfies

$$b_{k, H_0} = \left( \frac{\Gamma(H_0 - 1/2)}{\sqrt{2\pi}} \right)^k a_{k, H_0}. \quad (4.2.12)$$

In particular, the Hermite process of order  $k$  is standard when

$$b_{k, H_0} = \left( \frac{(k(H_0 - 1) + 1)(2k(H_0 - 1) + 1)}{k! [2\Gamma(2 - 2H_0) \sin((H_0 - \frac{1}{2})\pi)]^k} \right)^{1/2}. \quad (4.2.13)$$

*Proof* The equality in distribution (4.2.11) and the relation (4.2.12) between the normalizing constants follows from (B.2.16) in Appendix B.2 and Proposition 4.2.3, since the kernels in the right-hand sides of (4.2.4) and (4.2.12) are related through the Fourier transform. Note also from Proposition 4.2.3 that the Fourier transform of the kernel in (4.2.4) produces complex exponentials

$$\prod_{j=1}^k e^{-i \operatorname{sign}(x_j) \frac{(H_0 - 1/2)\pi}{2}},$$

which do not appear in the right-hand side of (4.2.11). The exponentials can be omitted by the formula (B.2.15) for the change of variables found in Appendix B. Finally, the form (4.2.13) of the normalizing constant follows from (4.2.12) and (4.2.8).  $\square$

### 4.2.3 Integral Representation on an Interval

The representations (4.2.4) and (4.2.11) of Hermite processes are on the real line. There is also an integral representation of Hermite processes on an interval. We continue working with the parametrization  $H_0$  only.

**Proposition 4.2.5** *The Hermite process of order  $k$  can be represented as*

$$\{Z_H^{(k)}(t)\}_{t \geq 0} \stackrel{d}{=} \left\{ d_{k, H_0} \int_{[0, t]^k}^{\prime} \left\{ \prod_{j=1}^k v_j^{\frac{1}{2} - H_0} \int_0^t s^{k(H_0 - \frac{1}{2})} \prod_{j=1}^k (s - v_j)_+^{H_0 - \frac{3}{2}} ds \right\} B(dv_1) \dots B(dv_k) \right\}_{t \geq 0}, \quad (4.2.14)$$

where  $B(dv)$  is a Gaussian random measure on  $\mathbb{R}$  with control measure  $dv$ . The normalizing constant  $d_{k, H_0}$  satisfies

$$d_{k, H_0} = a_{k, H_0}. \quad (4.2.15)$$

**Remark 4.2.6** When  $k = 1$ , the expression (4.2.14) provides an integral representation on an interval of FBM with  $H_0 \in (1/2, 1)$ . This representation of FBM is extended to  $H_0 \in (0, 1/2)$  in Chapters 6 and 7, and plays there a key role in the development of a stochastic calculus with respect to FBM.

The proof of Proposition 4.2.5 involves a number of auxiliary results, some of which have their own interest. We provide the proof next without using the usual delimiters PROOF and  $\square$  for the beginning and the end of the proof.

It is instructive to explain first the basic idea of the proof in informal terms. Interchanging the two integrals in the representation (4.2.4) and using Proposition 4.1.2, (a), we have

$$\begin{aligned} & \int_{\mathbb{R}^k}^{\prime} \left\{ \int_0^t \prod_{j=1}^k (s - u_j)_+^{H_0 - \frac{3}{2}} ds \right\} B(du_1) \dots dB(du_k) \\ &= \int_0^t \left\{ \int_{\mathbb{R}^k}^{\prime} \prod_{j=1}^k (s - u_j)_+^{H_0 - \frac{3}{2}} B(du_1) \dots dB(du_k) \right\} ds \\ &= \int_0^t H_k \left( \int_{\mathbb{R}} (s - u)_+^{H_0 - \frac{3}{2}} B(du) \right) ds. \end{aligned} \quad (4.2.16)$$

By finding a representation of a Gaussian process  $\int_{\mathbb{R}} (s - u)_+^{H_0 - \frac{3}{2}} B(du)$  on an interval and repeating the argument (4.2.16) backwards leads to the representation (4.2.14) of Hermite processes on an interval.

The Gaussian process  $\int_{\mathbb{R}} (s - u)_+^{H_0 - \frac{3}{2}} B(du)$  in (4.2.16) is not well-defined because  $\int_{\mathbb{R}} (s - u)_+^{2H_0 - 3} du = \infty$ . To be able to proceed as in (4.2.16), we use the idea of *regularization*. For  $\epsilon > 0$ , let

$$Z_{H, \epsilon}^{(k)}(t) = \int_{\mathbb{R}^k}^{\prime} \left\{ \int_0^t \prod_{j=1}^k (s - u_j)_+^{H_0 - \frac{3}{2}} 1_{\{s - u_j > \epsilon\}} ds \right\} B(du_1) \dots B(du_k) \quad (4.2.17)$$

be a regularized version of  $Z_H^{(k)}(t)$ . By using (B.2.25) in Appendix B.2 and Proposition 4.1.2, (a), we have

$$\begin{aligned}
Z_{H,\epsilon}^{(k)}(t) &= \int_0^t \left\{ \int_{\mathbb{R}^k} \prod_{j=1}^k (s - u_j)_+^{H_0 - \frac{3}{2}} 1_{\{s - u_j > \epsilon\}} B(du_1) \dots B(du_k) \right\} ds \\
&= \int_0^t \|\dot{B}_{H_0,\epsilon}(s)\|_2^k H_k \left( \frac{\dot{B}_{H_0,\epsilon}(s)}{\|\dot{B}_{H_0,\epsilon}(s)\|_2} \right) ds,
\end{aligned} \tag{4.2.18}$$

where  $\|X\|_2 = (\mathbb{E}X^2)^{1/2}$  for a random variable  $X$ , and

$$\dot{B}_{H_0,\epsilon}(s) = \int_{\mathbb{R}} (s - u)_+^{H_0 - \frac{3}{2}} 1_{\{s - u > \epsilon\}} B(du) \tag{4.2.19}$$

is a regularized fractional Gaussian noise.

We would like to have now a representation of the regularized Gaussian process  $\dot{B}_{H_0,\epsilon}$  on an interval. Such a representation, however, is not expected to have a simple analytic form. We shall therefore provide a representation of  $\dot{B}_{H_0,\epsilon}$  on the positive half-axis. This will lead to a representation of Hermite processes on the positive half-axis. We shall then regularize it in a different way in order to derive the representation (4.2.14) on a finite interval.

### Representation on the Positive Half-Axis

In the next auxiliary result, we establish an equivalent representation of  $\dot{B}_{H_0,\epsilon}$  on the positive-half axis.

**Lemma 4.2.7** *Let  $H_0 < 1$ . Then,*

$$\{\dot{B}_{H_0,\epsilon}(s)\}_{s \geq 0} \stackrel{d}{=} \{\dot{X}_{H_0,\epsilon}(s)\}_{s \geq 0}, \tag{4.2.20}$$

where

$$\dot{X}_{H_0,\epsilon}(s) = \int_0^\infty z^{\frac{1}{2} - H_0} (1 - sz)^{H_0 - \frac{3}{2}} 1_{\{z < 1/(\epsilon + s)\}} B(dz) \tag{4.2.21}$$

and  $B(dz)$  is a Gaussian random measure with control measure  $dz$ .

*Proof* The stationary Gaussian process  $\dot{B}_{H_0,\epsilon}(s)$  has the following covariance. For  $0 \leq s_1 < s_2$ ,

$$I := \mathbb{E} \dot{B}_{H_0,\epsilon}(s_2) \dot{B}_{H_0,\epsilon}(s_1) = \int_{\mathbb{R}} (s_2 - y)_+^{H_0 - \frac{3}{2}} (s_1 - y)_+^{H_0 - \frac{3}{2}} 1_{\{s_1 - y > \epsilon\}} dy.$$

The proof follows from a series of changes of variables. Some of them are rather delicate. By making the change of variables  $s_1 - y = z(s_2 - s_1)$ , we get that

$$I = (s_2 - s_1)^{2H_0 - 2} \int_0^\infty z^{H_0 - \frac{3}{2}} (1 + z)^{H_0 - \frac{3}{2}} 1_{\{z > \epsilon/(s_2 - s_1)\}} dz.$$

By making the change of variables  $z = (1 - u)/u$ , we obtain that

$$I = (s_2 - s_1)^{2H_0 - 2} \int_0^1 u^{1 - 2H_0} (1 - u)^{H_0 - \frac{3}{2}} 1_{\{u < (s_2 - s_1)/(\epsilon + s_2 - s_1)\}} du.$$

A further change of variables  $u = (s_2 - s_1)w/(s_1(s_2 - w))$  leads to

$$I = s_1^{H_0 - \frac{1}{2}} s_2^{H_0 - \frac{1}{2}} \int_0^{s_1} w^{1 - 2H_0} (s_2 - w)^{H_0 - \frac{3}{2}} (s_1 - w)^{H_0 - \frac{3}{2}} 1_{\{w < (s_1 s_2)/(\epsilon + s_2)\}} dw.$$



Finally, the change of variables  $w = s_1 s_2 z$  yields

$$I = \int_0^\infty z^{1-2H_0} (1 - s_2 z)^{H_0 - \frac{3}{2}} (1 - s_1 z)^{H_0 - \frac{3}{2}} 1_{\{z < 1/(\epsilon + s_1)\}} 1_{\{z < 1/(\epsilon + s_2)\}} dz,$$

which is  $\mathbb{E} \dot{X}_{H_0, \epsilon}(s_2) \dot{X}_{H_0, \epsilon}(s_1)$ . □

Lemma 4.2.7 and the relation (4.2.18) yield

$$\{Z_{H, \epsilon}^{(k)}(t)\}_{t \geq 0} \stackrel{d}{=} \left\{ \int_0^t \|\dot{X}_{H_0, \epsilon}(s)\|_2^k H_k \left( \frac{\dot{X}_{H_0, \epsilon}(s)}{\|\dot{X}_{H_0, \epsilon}(s)\|_2} \right) ds \right\}_{t \geq 0}. \quad (4.2.22)$$

Repeating the argument in (4.2.17) and (4.2.18) gives

$$\{Z_{H, \epsilon}^{(k)}(t)\}_{t \geq 0} \stackrel{d}{=} \{X_{H_0, \epsilon}^{(k)}(t)\}_{t \geq 0}, \quad (4.2.23)$$

where

$$X_{H_0, \epsilon}^{(k)}(t) = \int_{[0, \infty)^k}' \left\{ \int_0^t \prod_{j=1}^k z_j^{\frac{1}{2} - H_0} (1 - s z_j)^{H_0 - \frac{3}{2}} 1_{\{z_j < 1/(\epsilon + s)\}} ds \right\} B(dz_1) \dots B(dz_k). \quad (4.2.24)$$

Letting  $\epsilon \rightarrow 0$  in (4.2.23) yields the following integral representations of Hermite processes on the positive half-axis.

**Proposition 4.2.8** *One has*

$$\{Z_{H_0}^{(k)}(t)\}_{t \geq 0} \stackrel{d}{=} \{X_{H_0}^{(k)}(t)\}_{t \geq 0}, \quad (4.2.25)$$

where

$$X_{H_0}^{(k)}(t) = \int_{[0, \infty)^k}' \left\{ \int_0^t \prod_{j=1}^k z_j^{\frac{1}{2} - H_0} (1 - s z_j)_+^{H_0 - \frac{3}{2}} ds \right\} B(dz_1) \dots B(dz_k) \quad (4.2.26)$$

and  $B(dz)$  is a Gaussian random measure on  $\mathbb{R}$  with control measure  $dz$ .

*Proof* Relation (4.2.25) is obtained by letting  $\epsilon \rightarrow 0$  in (4.2.23), and using the fact that for fixed  $t$ ,  $Z_{H, \epsilon}^{(k)}(t)$  and  $X_{H_0, \epsilon}^{(k)}(t)$  converge in the  $L^2(\Omega)$ -sense to the corresponding limit random variables  $Z_H^{(k)}(t)$  and  $X_{H_0}^{(k)}(t)$ . The latter convergence is a consequence of the fact that the corresponding deterministic integrand functions converge in  $L^2(\mathbb{R}^k)$ . For example, in the case of  $Z_{H, \epsilon}^{(k)}(t)$  and  $Z_H^{(k)}(t)$ , one needs to consider

$$\begin{aligned} & \int_{\mathbb{R}^k} \left( \int_0^t \prod_{j=1}^k (s - u_j)_+^{H_0 - \frac{3}{2}} 1_{\{s - u_j > \epsilon\}} ds - \int_0^t \prod_{j=1}^k (s - u_j)_+^{H_0 - \frac{3}{2}} ds \right)^2 du_1 \dots du_k \\ &= \int_{\mathbb{R}^k} \left( \int_0^t \prod_{j=1}^k (s - u_j)_+^{H_0 - \frac{3}{2}} 1_{\{s - u_j \leq \epsilon\}} ds \right)^2 du_1 \dots du_k. \end{aligned}$$

As  $\epsilon \rightarrow 0$ , this last integral converges to 0 by the dominated convergence theorem and the fact that the same integral without the indicator functions is finite, by (4.2.2). The same argument can be repeated for  $X_{H_0, \epsilon}^{(k)}(t)$  and  $X_{H_0}^{(k)}(t)$ . In this case, note in addition that

$$\begin{aligned} & \int_{[0, \infty)^k} \left( \int_0^t \prod_{j=1}^k z_j^{\frac{1}{2}-H_0} (1-sz_j)_+^{H_0-\frac{3}{2}} ds \right)^2 dz_1 \dots dz_k \\ &= \lim_{\epsilon \rightarrow 0} \int_{[0, \infty)^k} \left( \int_0^t \prod_{j=1}^k z_j^{\frac{1}{2}-H_0} (1-sz_j)_+^{H_0-\frac{3}{2}} 1_{\{z_j < 1/(\epsilon+s)\}} ds \right)^2 dz_1 \dots dz_k \\ &= \lim_{\epsilon \rightarrow 0} \int_{\mathbb{R}^k} \left( \int_0^t \prod_{j=1}^k (s-u_j)_+^{H_0-\frac{3}{2}} 1_{\{s-u_j > \epsilon\}} ds \right)^2 du_1 \dots du_k \\ &= \int_{\mathbb{R}^k} \left( \int_0^t \prod_{j=1}^k (s-u_j)_+^{H_0-\frac{3}{2}} ds \right)^2 du_1 \dots du_k < \infty, \end{aligned}$$

where the second equality above is a consequence of (4.2.23), so that  $X_{H_0}^{(k)}(t)$  is well-defined.  $\square$

### Representation on an Interval

The next step is to regularize  $X_{H_0}^{(k)}$  but using a regularization different from (4.2.24). For  $\epsilon > 0$ , consider

$$W_{H_0, \epsilon}^{(k)}(t) = \int_{[0, \infty)^k} \left\{ \int_0^t \prod_{j=1}^k z_j^{\frac{1}{2}-H_0} (1-sz_j)^{H_0-\frac{3}{2}} 1_{\{z_j < (1-\epsilon)/s\}} ds \right\} B(dz_1) \dots B(dz_k). \quad (4.2.27)$$

As  $\epsilon \rightarrow 0$ ,  $W_{H_0, \epsilon}^{(k)}(t)$  converges to  $X_{H_0}^{(k)}(t)$ . By arguing as in (4.2.17) and (4.2.18), write

$$W_{H_0, \epsilon}^{(k)}(t) = \int_0^t \|\dot{W}_{H_0, \epsilon}(s)\|_2^k H_k \left( \frac{\dot{W}_{H_0, \epsilon}(s)}{\|\dot{W}_{H_0, \epsilon}(s)\|_2} \right) ds, \quad (4.2.28)$$

where

$$\dot{W}_{H_0, \epsilon}(s) = \int_0^\infty z^{\frac{1}{2}-H_0} (1-sz)^{H_0-\frac{3}{2}} 1_{\{z < (1-\epsilon)/s\}} B(dz). \quad (4.2.29)$$

Whereas in (4.2.24) the regularization involved  $1_{\{z < 1/(\epsilon+s)\}}$ , here the regularization involves  $1_{\{z < (1-\epsilon)/s\}}$ . In the next result, we provide an integral representation of  $\dot{W}_{H_0, \epsilon}$  on an interval.

**Proposition 4.2.9** *Let  $H_0 < 1$ . Then,*

$$\{\dot{W}_{H_0, \epsilon}(s)\}_{s \geq 0} \stackrel{d}{=} \{\dot{Y}_{H_0, \epsilon}(s)\}_{s \geq 0}, \quad (4.2.30)$$

where

$$\dot{Y}_{H_0, \epsilon}(s) = s^{H_0-\frac{1}{2}} \int_0^\infty v^{\frac{1}{2}-H_0} (s-v)^{H_0-\frac{3}{2}} 1_{\{v < s/(1-\epsilon)\}} B(dv) \quad (4.2.31)$$

and  $B(dv)$  is a Gaussian random measure with control measure  $dv$ .

*Proof* Let  $0 \leq s_1 < s_2$ . By the change of variables  $z = x/(s_1 s_2)$ ,

$$\begin{aligned} \mathbb{E} \dot{W}_{H_0, \epsilon}(s_2) \dot{W}_{H_0, \epsilon}(s_1) &= \int_0^\infty z^{1-2H_0} (1 - s_2 z)^{H_0 - \frac{3}{2}} (1 - s_1 z)^{H_0 - \frac{3}{2}} \mathbf{1}_{\{z < (1-\epsilon)/s_2\}} \mathbf{1}_{\{z < (1-\epsilon)/s_1\}} dz \\ &= (s_1 s_2)^{H_0 - \frac{1}{2}} \int_0^\infty x^{1-2H_0} (s_2 - x)^{H_0 - \frac{3}{2}} (s_1 - x)^{H_0 - \frac{3}{2}} \mathbf{1}_{\{x < s_2(1-\epsilon)\}} \mathbf{1}_{\{x < s_1(1-\epsilon)\}} dx, \end{aligned}$$

which is also  $\mathbb{E} \dot{Y}_{H_0, \epsilon}(s_2) \dot{Y}_{H_0, \epsilon}(s_1)$ .  $\square$

Proposition 4.2.9 and the relation (4.2.28) yield

$$\{W_{H_0, \epsilon}^{(k)}(t)\}_{t \geq 0} \stackrel{d}{=} \{Y_{H_0, \epsilon}^{(k)}(t)\}_{t \geq 0}, \quad (4.2.32)$$

where

$$\begin{aligned} Y_{H_0, \epsilon}^{(k)}(t) &= \int_{[0, \infty)^k}' \left\{ \prod_{j=1}^k v_j^{\frac{1}{2} - H_0} \int_0^t s^{k(H_0 - \frac{1}{2})} \prod_{j=1}^k (s - v_j)^{H_0 - \frac{3}{2}} \mathbf{1}_{\{v_j < s(1-\epsilon)\}} ds \right\} B(dv_1) \dots B(dv_k). \end{aligned} \quad (4.2.33)$$

Letting  $\epsilon \rightarrow 0$  and using Proposition 4.2.8, we obtain the integral representation (4.2.14) of Hermite processes on an interval. The proof is analogous to that of Proposition 4.2.8, and is omitted.

#### 4.2.4 Summary

For the convenience of the reader, we gather the time, spectral, half-axis and finite interval representations of the Hermite processes in the following theorem.

**Theorem 4.2.10** *Let  $H$  and  $H_0$  satisfy (4.2.1) and (4.2.2), respectively. Let also  $B(du)$  be a Gaussian random measure on  $\mathbb{R}$  with control measure  $du$ , and  $\widehat{B}(dx)$  be an Hermitian Gaussian random measure on  $\mathbb{R}$  with control measure  $dx$ . Then, the Hermite process  $Z_H^{(k)} = \{Z_H^{(k)}(t)\}_{t \in \mathbb{R}}$  has the following integral representations in the sense of equality of the finite-dimensional distributions:*

(a) *Time domain representation: for  $t \in \mathbb{R}$ ,*

$$a_{k, H_0} \int_{\mathbb{R}^k}' \left\{ \int_0^t \prod_{j=1}^k (s - u_j)_+^{H_0 - \frac{3}{2}} ds \right\} B(du_1) \dots dB(du_k). \quad (4.2.34)$$

(b) *Spectral domain representation: for  $t \in \mathbb{R}$ ,*

$$b_{k, H_0} \int_{\mathbb{R}^k}'' \frac{e^{it(x_1 + \dots + x_k)} - 1}{i(x_1 + \dots + x_k)} \prod_{j=1}^k |x_j|^{\frac{1}{2} - H_0} \widehat{B}(dx_1) \dots \widehat{B}(dx_k). \quad (4.2.35)$$

(c) *Positive half-axis representation: for  $t \geq 0$ ,*

$$c_{k,H_0} \int_{[0,\infty)^k}' \left\{ \int_0^t \prod_{j=1}^k z_j^{\frac{1}{2}-H_0} (1-sz_j)_+^{H_0-\frac{3}{2}} ds \right\} B(dz_1) \dots B(dz_k). \quad (4.2.36)$$

(d) *Finite interval representation: for  $t \geq 0$ ,*

$$d_{k,H_0} \int_{[0,t]^k}' \left\{ \prod_{j=1}^k v_j^{\frac{1}{2}-H_0} \int_0^t s^{k(H_0-\frac{1}{2})} \prod_{j=1}^k (s-v_j)_+^{H_0-\frac{3}{2}} ds \right\} B(dv_1) \dots B(dv_k). \quad (4.2.37)$$

If the Hermite process  $Z_H^{(k)}$  is standard so that  $\mathbb{E}(Z_H^{(k)}(1))^2 = 1$ , then

$$a_{k,H_0} = c_{k,H_0} = d_{k,H_0} = \left( \frac{(1-k(1-H_0))(1-2k(1-H_0))}{k!(B(H_0-\frac{1}{2}, 2-2H_0))^k} \right)^{1/2} \quad (4.2.38)$$

$$= \left( \frac{(k(H_0-1)+1)(2k(H_0-1)+1)\Gamma(3/2-H_0)^k}{k![\Gamma(H_0-1/2)\Gamma(2-2H_0)]^k} \right)^{1/2} \quad (4.2.39)$$

and

$$b_{k,H_0} = \left( \frac{(k(H_0-1)+1)(2k(H_0-1)+1)}{k![\Gamma(2-2H_0)\sin((H_0-\frac{1}{2})\pi)]^k} \right)^{1/2}. \quad (4.2.40)$$

*Proof* This follows from Definition 4.2.1, and Propositions 4.2.2, 4.2.4, 4.2.5 and 4.2.8.  $\square$

It is sometimes useful to have the expressions in Theorem 4.2.10 expressed in terms of  $H$  instead of  $H_0$ . Since  $H$  and  $H_0$  are related through (4.2.2), one has:

**Corollary 4.2.11** *The Hermite process  $Z_H^{(k)} = \{Z_H^{(k)}(t)\}_{t \in \mathbb{R}}$  has the following integral representations in the sense of equality of the finite-dimensional distributions:*

(a) *Time domain representation: for  $t \in \mathbb{R}$ ,*

$$A_{k,H} \int_{\mathbb{R}^k}' \left\{ \int_0^t \prod_{j=1}^k (s-u_j)_+^{-\left(\frac{1}{2}+\frac{1-H}{k}\right)} ds \right\} B(du_1) \dots dB(du_k). \quad (4.2.41)$$

(b) *Spectral domain representation: for  $t \in \mathbb{R}$ ,*

$$B_{k,H} \int_{\mathbb{R}^k}'' \frac{e^{it(x_1+\dots+x_k)} - 1}{i(x_1+\dots+x_k)} \prod_{j=1}^k |x_j|^{\frac{1-H}{k}-\frac{1}{2}} \widehat{B}(dx_1) \dots \widehat{B}(dx_k). \quad (4.2.42)$$

(c) *Positive half-axis representation: for  $t \geq 0$ ,*

$$C_{k,H} \int_{[0,\infty)^k}' \left\{ \int_0^t \prod_{j=1}^k z_j^{\frac{1-H}{k}-\frac{1}{2}} (1-sz_j)_+^{-\left(\frac{1}{2}+\frac{1-H}{k}\right)} ds \right\} B(dz_1) \dots B(dz_k). \quad (4.2.43)$$

(d) *Finite interval representation*: for  $t \geq 0$ ,

$$D_{k,H} \int_{[0,t]^k}' \left\{ \prod_{j=1}^k v_j^{\frac{1-H}{k}-\frac{1}{2}} \int_0^t s^{\frac{k}{2}-(1-H)} \prod_{j=1}^k (s-v_j)_+^{-(\frac{1}{2}+\frac{1-H}{k})} ds \right\} B(dv_1) \dots B(dv_k). \quad (4.2.44)$$

If  $\mathbb{E}(Z_H^{(k)}(1))^2 = 1$ , then

$$a_{k,H_0} = A_{k,H}, \quad b_{k,H_0} = B_{k,H}, \quad c_{k,H_0} = C_{k,H}, \quad d_{k,H_0} = D_{k,H},$$

where

$$A_{k,H} = C_{k,H} = D_{k,H} = \left( \frac{H(2H-1)\Gamma(\frac{1}{2} + \frac{1-H}{k})^k}{k! [\Gamma(\frac{1}{2} - \frac{1-H}{k})\Gamma(\frac{2(1-H)}{k})]^k} \right)^{1/2} \quad (4.2.45)$$

and<sup>4</sup>

$$B_{k,H} = \left( \frac{H(2H-1)}{k! [2\Gamma(\frac{2(1-H)}{k}) \sin((\frac{1}{2} - \frac{1-H}{k})\pi)]^k} \right)^{1/2}. \quad (4.2.46)$$

### 4.3 Moments, Cumulants and Diagram Formulae for Multiple Integrals

We will provide next formulae for the moments of multiple integrals and Hermite polynomials of Gaussian variables. These results are expressed in terms of diagrams and related notions, which we recall first.

#### 4.3.1 Diagram Formulae

For a nonempty set  $b$ , let  $\mathcal{P}(b)$  denote the set of its partitions. For each pair  $i, j \in b$  and for each  $\pi \in \mathcal{P}(b)$ , we write

$$i \sim_{\pi} j$$

whenever  $i$  and  $j$  belong to the same block of  $\pi$ . For example, if  $b = \{1, 2, 3, 4\}$  and  $\pi = \{\{1, 2, 3\}, \{4\}\}$ , then  $1 \sim_{\pi} 2$ .

**Definition 4.3.1** A *diagram* is a graphical representation of a pair of partitions  $\pi, \sigma \in \mathcal{P}(b)$ . It is obtained as follows. Let  $\pi = \{b_1, \dots, b_p\}$  and  $\sigma = \{t_1, \dots, t_q\}$  represent the blocks of the two partitions. Then:

1. Order the elements of each block  $b_i$ , for  $i = 1, \dots, p$ .
2. Associate with each block  $b_i \in \pi$  a *row* of vertices (represented as dots), in such a way that the  $j$ th vertex of the  $i$ th row corresponds to the  $j$ th element of the block  $b_i$ .
3. For every  $a = 1, \dots, q$ , draw a *closed curve* around the vertices corresponding to the elements of the block  $t_a \in \sigma$ .

We will denote by  $\Gamma(\pi, \sigma)$  the diagram of a pair of partitions  $\pi, \sigma$ . Note that the rows of the diagram indicate the blocks in  $\pi$  and the curves indicate the blocks in  $\sigma$ .

<sup>4</sup> There is a typo in the expression of (1.20) of Pipiras and Taqqu [819], namely an extra factor  $k$ .

**Definition 4.3.2** A *non-flat* diagram  $\Gamma(\pi, \sigma)$  is such that the closed curves defining the blocks of  $\sigma$  cannot join two vertices in the same row (thus having a flat or horizontal portion). We say that the diagram  $\Gamma(\pi, \sigma)$  is *Gaussian*, whenever every block of  $\sigma$  contains exactly two elements.

When a diagram is Gaussian, one usually represents the blocks of  $\sigma$  not by closed curves, but by segments connecting two vertices.

**Example 4.3.3** (*Non-flat (Gaussian) diagrams*) Figure 4.1, (a), provides an example of a non-flat diagram  $\Gamma(\pi, \sigma)$ , where  $b = \{1, 2, 3, 4, 5, 6, 7\}$ ,  $\pi = \{\{1, 2, 3\}, \{4\}, \{5, 6, 7\}\}$  and  $\sigma = \{\{1, 4, 5\}, \{2, 7\}, \{3, 6\}\}$ .

Figure 4.1, (b), provides an example of a non-flat Gaussian diagram  $\Gamma(\pi, \sigma)$ , where  $b = \{1, 2, 3, 4, 5, 6\}$ ,  $\pi = \{\{1, 2, 3\}, \{4\}, \{5, 6\}\}$  and  $\sigma = \{\{1, 4\}, \{2, 5\}, \{3, 6\}\}$ .

Let  $n_1, \dots, n_p \geq 1$ . Denote

$$n = n_1 + \dots + n_p, \quad [n] = \{1, \dots, n\}, \quad (4.3.1)$$

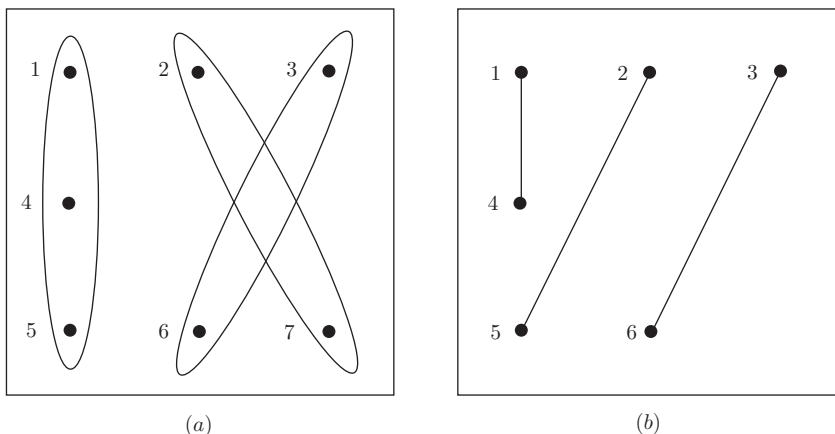
and consider the following special partition of the set  $[n]$ :

$$\pi^* = \{\{1, \dots, n_1\}, \{n_1 + 1, \dots, n_1 + n_2\}, \dots, \{n_1 + \dots + n_{p-1} + 1, \dots, n\}\} \in \mathcal{P}([n]). \quad (4.3.2)$$

Now define

$$\mathcal{M}_2^0([n], \pi^*) = \{\sigma \in \mathcal{P}([n]) : \Gamma(\pi^*, \sigma) \text{ is non-flat and Gaussian}\}.$$

The moment formula for multiple integrals, called the *diagram formula*, is the following. See, for example, Peccati and Taqqu [797], Corollary 7.3.1, p. 133.



**Figure 4.1** (a) A non-flat diagram. (b) A non-flat Gaussian diagram.

**Proposition 4.3.4** (Diagram formula.) For symmetric  $f_i \in L^2(E^{n_i}, m^{n_i})$ ,  $i = 1, \dots, p$ , and non-atomic  $m$ ,

$$\mathbb{E}(I_{n_1}(f_1) \dots I_{n_p}(f_p)) = \sum_{\sigma \in \mathcal{M}_2^0([n], \pi^*)} \int_{E^{n/2}} f_{\sigma, p} dm^{n/2}, \quad (4.3.3)$$

where, for every  $\sigma \in \mathcal{M}_2^0([n], \pi^*)$ , the function  $f_{\sigma, p}$  of  $n/2$  variables, is obtained by identifying the variables  $u_i$  and  $u_j$  in the argument of

$$f_1(u_1, \dots, u_{n_1}) f_2(u_{n_1+1}, \dots, u_{n_2}) \dots f_p(u_{n_1+\dots+n_{p-1}+1}, \dots, u_n)$$

if and only if  $i \sim_{\sigma} j$ . If  $\mathcal{M}_2^0([n], \pi^*) = \emptyset$  (in particular, if  $n$  is odd), the right-hand side of (4.3.3) is interpreted as 0.

**Example 4.3.5** (Diagram formula) Let us evaluate  $\mathbb{E}I_3(f_1)I_1(f_2)I_2(f_3)$  where  $f_1$ ,  $f_2$  and  $f_3$  are respectively symmetric functions of 3, 1 and 2 variables. The function

$$f_1(u_1, u_2, u_3) f_2(u_4) f_3(u_5, u_6)$$

is a function of 6 variables. In this case,  $b = \{1, 2, 3, 4, 5, 6\}$  and  $\pi^* = \{\{1, 2, 3\}, \{4\}, \{5, 6\}\}$ . One of the partitions in  $\mathcal{M}_2^0([6], \pi^*)$  is  $\sigma = \{\{1, 4\}, \{2, 5\}, \{3, 6\}\}$ , with the corresponding diagram  $\Gamma(\pi^*, \sigma)$  depicted in Figure 4.1, (b). This diagram is non-flat and Gaussian. It is associated to the summand

$$\int_{E^3} f_1(u_1, u_2, u_3) f_2(u_1) f_3(u_2, u_3) m(du_1) m(du_2) m(du_3) \quad (4.3.4)$$

in the sum on the right-hand side of (4.3.3), where, according to  $\sigma$ , we identified  $u_4$  with  $u_1$ ,  $u_5$  with  $u_2$  and  $u_6$  with  $u_3$ . To obtain the other summands, one needs to find all the partitions  $\sigma$  in  $\mathcal{M}_2^0([6], \pi^*)$ .<sup>5</sup> One gets

$$\begin{aligned} \mathbb{E}I_3(f_1)I_1(f_2)I_2(f_3) &= \int_{E^3} \left( f_1(u_1, u_2, u_3) f_2(u_3) f_3(u_2, u_1) \right. \\ &\quad + f_1(u_1, u_2, u_3) f_2(u_3) f_3(u_1, u_2) + f_1(u_1, u_2, u_3) f_2(u_2) f_3(u_3, u_1) \\ &\quad + f_1(u_1, u_2, u_3) f_2(u_1) f_3(u_3, u_2) + f_1(u_1, u_2, u_3) f_2(u_2) f_3(u_1, u_3) \\ &\quad \left. + f_1(u_1, u_2, u_3) f_2(u_1) f_3(u_2, u_3) \right) m(du_1) m(du_2) m(du_3) \end{aligned} \quad (4.3.5)$$

$$= 6 \int_{E^3} f_1(u_1, u_2, u_3) f_2(u_3) f_3(u_2, u_1) m(du_1) m(du_2) m(du_3). \quad (4.3.6)$$

The last line is a consequence of the symmetry of the functions. We will see below that we can obtain the result immediately by using “multigraphs.”

<sup>5</sup> This can be done by applying the Mathematica program described in Peccati and Taqqu [797]. One executes the command `MZeroSetsEqualTwo[{3, 1, 2}]`, where the entry  $\{3, 1, 2\}$  defines the block sizes of  $\pi^*$ . The output indicates that there are 6 partitions  $\sigma$  in  $\mathcal{M}_2^0([6], \pi^*)$  and lists them as  $\{\{1, 6\}, \{2, 5\}, \{3, 4\}\}$ ,  $\{\{1, 5\}, \{2, 6\}, \{3, 4\}\}$ ,  $\{\{1, 6\}, \{2, 4\}, \{3, 5\}\}$ ,  $\{\{1, 4\}, \{2, 6\}, \{3, 5\}\}$ ,  $\{\{1, 5\}, \{2, 4\}, \{3, 6\}\}$ ,  $\{\{1, 4\}, \{2, 5\}, \{3, 6\}\}$ . This shows how to equalize the variables. For example,  $\{\{1, 6\}, \{2, 5\}, \{3, 4\}\}$  means “set  $u_1 = u_6, u_2 = u_5, u_3 = u_4$ .”

Note that the diagram formula (4.3.3) concerns multiple integrals in the time domain. There is an analogous formula for multiple integrals  $\widehat{I}_n(f_n)$  in the spectral domain. In this case, suppose  $E = \mathbb{R}$ ,  $m(du) = m(-du)$  and the symmetric functions  $f_n$  are Hermitian, namely,  $\overline{f_n(u)} = f_n(-u)$ . Then, in Proposition 4.3.4, when  $i \sim_\pi j$ , instead of identifying  $u_i$  and  $u_j$ , that is, setting  $u_j = u_i$ , one sets  $u_j = -u_i$  (see also Surgailis [937], Proposition 5.1).

**Example 4.3.6** (*Diagram formula in the spectral domain*) Suppose we want to evaluate  $\mathbb{E}\widehat{I}_3(f_1)\widehat{I}_1(f_2)\widehat{I}_2(f_3)$  where the functions  $f_1$ ,  $f_2$  and  $f_3$  are as in Example 4.3.5 but Hermitian and suppose  $m(du) = m(-du)$ . Then (4.3.5) holds with  $\mathbb{E}\widehat{I}_3(f_1)\widehat{I}_1(f_2)\widehat{I}_2(f_3)$  on its left-hand side but with the arguments of  $f_2$  and  $f_3$  on its right-hand side having changed signs. Hence, (4.3.4) becomes

$$\int_{E^3} f_1(u_1, u_2, u_3) f_2(-u_1) f_3(-u_2, -u_3) m(du_1) m(du_2) m(du_3).$$

Similarly, the equality (4.3.6) continues to hold after the same change as for the right-hand side of (4.3.5).

The formula (4.3.3) concerns the moments of multiple integrals. There is a similar formula for the cumulants of multiple integrals.

**Definition 4.3.7** For a vector of real-valued random variables  $(X_1, \dots, X_p)$  such that  $\mathbb{E}|X_j|^p < \infty$ ,  $j = 1, \dots, p$ , its *joint cumulant* is defined as

$$\chi(X_1, \dots, X_p) = (-i)^p \frac{\partial^p}{\partial z_1 \dots \partial z_p} \log \mathbb{E} \left( \exp \left\{ i \sum_{j=1}^p z_j X_j \right\} \right) \Big|_{z_1=\dots=z_p=0}. \quad (4.3.7)$$

The usual  $p$ th cumulant of a random variable  $X$  such that  $\mathbb{E}|X|^p < \infty$  is defined as

$$\chi_p(X) = \chi(\underbrace{X, \dots, X}_{p \text{ times}}) = (-i)^p \frac{\partial^p}{\partial z^p} \log \mathbb{E} \left( \exp \{ i z X \} \right) \Big|_{z=0}. \quad (4.3.8)$$

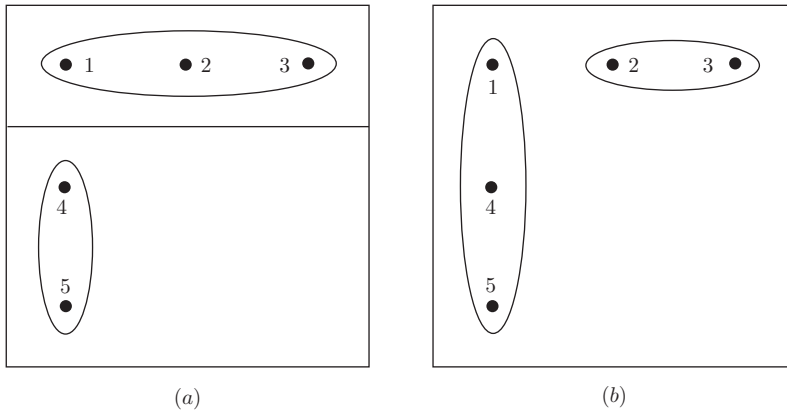
One has, for example,  $\chi_1(X) = \mathbb{E}X$ ,  $\chi_2(X) = \mathbb{E}X^2 - (\mathbb{E}X)^2$ ,  $\chi_3(X) = \mathbb{E}X^3 - 3\mathbb{E}X^2\mathbb{E}X + 2(\mathbb{E}X)^3$ .

To state the result for the cumulants of multiple integrals, we need another notion related to diagrams.

**Definition 4.3.8** The diagram  $\Gamma(\pi, \sigma)$ ,  $\pi, \sigma \in \mathcal{P}(b)$ , is *connected* if the only partition  $\rho \in \mathcal{P}(b)$  for which each block of  $\pi$  and each block of  $\sigma$  is contained in a block of  $\rho$ , is the maximal partition consisting of one block with all the elements of  $b$ .

**Example 4.3.9** (*(Non-)Connected diagrams*) Figure 4.2, (a), provides an example of a non-connected diagram with  $b = \{1, 2, 3, 4, 5\}$ ,  $\pi = \{\{1, 2, 3\}, \{4\}, \{5\}\}$  and  $\sigma = \{\{1, 2, 3\}, \{4, 5\}\}$ , for which there exists a non-maximal partition  $\rho = \{\{1, 2, 3\}, \{4, 5\}\}$ . The blocks  $\{1, 2, 3\}$  and  $\{4, 5\}$  of this  $\rho$  contain the blocks of both  $\pi$  and  $\sigma$ , and this  $\rho$  is not the maximal partition  $\{1, 2, 3, 4, 5\}$ . Therefore the diagram  $\Gamma(\pi, \sigma)$  is not connected.





**Figure 4.2** (a) Dividing a non-connected diagram. (b) A connected diagram.

Figure 4.2, (b), provides an example of a connected diagram with  $b = \{1, 2, 3, 4, 5\}$ ,  $\pi = \{\{1, 2, 3\}, \{4\}, \{5\}\}$  and  $\sigma = \{\{1, 4, 5\}, \{2, 3\}\}$ . The only partition  $\rho$  containing each block of  $\pi$  and of  $\sigma$  is the maximal partition  $\rho = b = \{1, 2, 3, 4, 5\}$ .

Here is an easy way to check geometrically if a diagram  $\Gamma(\pi, \sigma)$  is connected or not. It is connected if and only if the rows of the diagrams (the blocks of  $\pi$ ) cannot be divided into two subsets, each defining a separate diagram. Thus, in Figure 4.2, the diagram in (a) is not connected (the horizontal line divides it into two diagrams), but the diagram in (b) is connected.

Now define

$$\mathcal{M}_2([n], \pi^*) = \{\sigma \in \mathcal{P}([n]) : \Gamma(\pi^*, \sigma) \text{ is connected, non-flat and Gaussian}\}.$$

The formula for the cumulants of multiple integrals is the following (Peccati and Taqqu [797], Corollary 7.3.1, p. 133).

**Proposition 4.3.10** (Diagram formula.) For symmetric  $f_i \in L^2(E^{n_i}, m^{n_i})$ ,  $i = 1, \dots, p$ , and non-atomic  $m$ ,

$$\chi(I_{n_1}(f_1), \dots, I_{n_p}(f_p)) = \sum_{\sigma \in \mathcal{M}_2([n], \pi^*)} \int_{E^{n/2}} f_{\sigma, p} dm^{n/2}, \quad (4.3.9)$$

where  $f_{\sigma, p}$  is defined as in (4.3.3). If  $\mathcal{M}_2([n], \pi^*) = \emptyset$  (in particular, if  $n$  is odd), the right-hand side of (4.3.9) is interpreted as 0.

Since  $\mathcal{M}_2([n], \pi^*) \subset \mathcal{M}_2^0([n], \pi^*)$ , there are fewer summands in the cumulant formula (4.3.9) than in the moment formula (4.3.3).

### 4.3.2 Multigraphs

There can be many diagrams. It is possible to simplify the formulae (4.3.3) and (4.3.9) by using multigraphs instead.

**Definition 4.3.11** A *multigraph*  $A = (V, E)$  consists of a nonempty set of points (vertices)  $V$ , and a (possibly empty) set of lines (edges)  $E$ . A multigraph may possess multiple lines connecting any two points.<sup>6</sup> A *labeled* (point-labeled) multigraph has its points distinguished from each other. Two labeled multigraphs will be assumed identical if for all  $i$  and  $j$ , the number of lines joining the points  $i$  and  $j$  is the same in both multigraphs.

For  $p \geq 1$ , let  $\mathcal{A}_p$  be the set of all labeled multigraphs with  $p$  points, and whose lines do not form loops, that is, do not start and end at the same point.  $A \in \mathcal{A}_p$  can be described through its  $p$  points and its *set of lines* or *pair sequence*

$$\{(i_1, j_1), \dots, (i_q, j_q)\} \quad (4.3.10)$$

as follows:  $q$  denotes the number of lines of  $A$  and for each  $s = 1, \dots, q$ , we have  $i_s, j_s \in \{1, \dots, p\}$  with  $i_s \neq j_s$ , a pair  $(i_s, j_s)$  symbolizing the existence of a line that joins point  $i_s$  to point  $j_s$ . Identical pairs indicate the presence of multiple lines. For example, the pair sequence of the multigraph  $A_1$  in Figure 4.3 is  $\{(1, 2), (1, 2), (3, 4), (3, 4)\}$ . The requirement  $i_s \neq j_s$  for all  $s = 1, \dots, q$  indicates the absence of loops.

With each multigraph  $A \in \mathcal{A}_p$ , we also associate a *multiplicity number*  $g(A)$  defined as follows. Number each of the  $\binom{p}{2}$  possible pairs of points by  $u = 1, 2, \dots, \binom{p}{2}$ . Let  $v_u$  be the number of lines in  $A$  joining the pair of points numbered  $u$ . Define

$$g(A) = \prod_{u=1}^{\binom{p}{2}} \frac{1}{v_u!}, \quad (4.3.11)$$

with the convention  $0! = 1$ . If  $A$  has  $q$  lines, then obviously  $v_1 + v_2 + \dots + v_{\binom{p}{2}} = q$ .

**Definition 4.3.12** Given  $p \geq 1, n_1, \dots, n_p \geq 1$ , let  $\mathcal{A}(n_1, \dots, n_p)$  denote the set of all multigraphs  $A$  of  $\mathcal{A}_p$  whose  $p$  points have degrees  $n_1, \dots, n_p$  respectively. The *degree* of a point is the number of lines incident to that point; that is, the number of line with one of their endpoints at that point. For  $k \geq 1$ , let also  $\mathcal{A}_p(k)$  denote the set of all multigraphs  $A$  of  $\mathcal{A}_p$  such that all the points of  $A$  have degree  $k$ . These multigraphs are called *k-regular*.

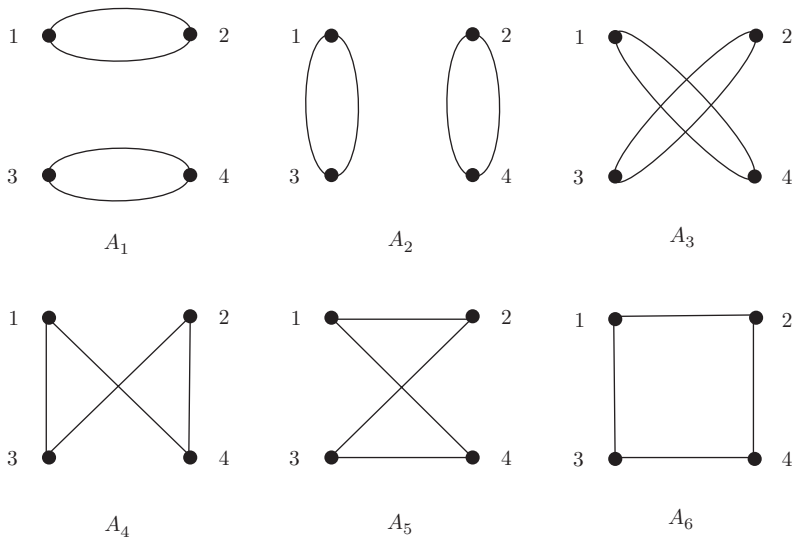
**Example 4.3.13** (*2-regular multigraphs*) Consider the set of  $p = 4$  points  $V = \{1, 2, 3, 4\}$ , and  $k = 2$ . Figure 4.3 shows all *2-regular* multigraphs, that is, all multigraphs of the set  $\mathcal{A}_4(2)$ , where each point of the set  $V$  has degree 2. There are six such multigraphs. The respective multiplicity numbers are

$$g(A_1) = g(A_2) = g(A_3) = \frac{1}{2!2!} = \frac{1}{4}, \quad g(A_4) = g(A_5) = g(A_6) = 1.$$

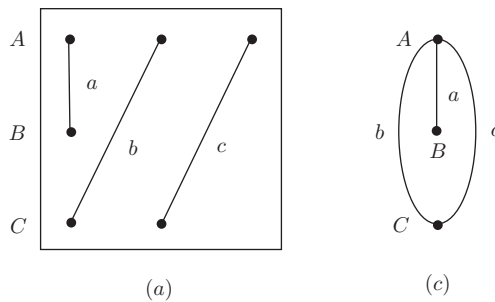
### 4.3.3 Relation Between Diagrams and Multigraphs

With  $n = n_1 + \dots + n_p$  and the special partition  $\pi^*$  in (4.3.2), a partition  $\sigma \in \mathcal{M}_2^0([n], \pi^*)$  or the corresponding non-flat, Gaussian diagram  $\Gamma(\pi^*, \sigma)$  can naturally be associated with a multigraph  $A \in \mathcal{A}(n_1, \dots, n_p)$ .

<sup>6</sup> “Multi” in “Multigraph” refers to this fact.



**Figure 4.3** The six multigraphs of the set  $\mathcal{A}_4(2)$ .



**Figure 4.4** Non-flat, Gaussian diagram in (a) and the multigraph (c) associated with them.

**Definition 4.3.14** We say that “a (non-flat, Gaussian) diagram  $\Gamma(\pi^*, \sigma)$  is associated with a multigraph  $A$ ”, or simply, that “ $\sigma$  is associated with the multigraph  $A$ ” if  $A$  is obtained from  $\sigma$  in the following way: Start with a diagram  $\Gamma(\pi^*, \sigma)$ . Suppose that  $\pi^*$  has  $p$  blocks and  $\sigma$  has  $q$  blocks corresponding to  $q$  segments. These  $p$  blocks will contribute  $p$  points to a multigraph  $A$  and these  $q$  segments will contribute  $q$  lines to  $A$ , as follows. Each segment in  $\Gamma(\pi^*, \sigma)$  connecting points in the  $i$ th block  $\{n_1 + \cdots + n_{i-1} + 1, \dots, n_1 + \cdots + n_i\}$  and the  $j$ th block  $\{n_1 + \cdots + n_{j-1} + 1, \dots, n_1 + \cdots + n_j\}$  of  $\pi^*$  contributes a line connecting points  $i$  and  $j$  in a multigraph  $A$ . Since there are  $n_i$  segments originating from the  $i$ th block  $\{n_1 + \cdots + n_{i-1} + 1, \dots, n_1 + \cdots + n_i\}$  of  $\pi^*$ , the point  $i$  of the multigraph  $A$  will have degree  $n_i$  and hence  $A$  will be a multigraph from  $\mathcal{A}(n_1, \dots, n_p)$ .

**Example 4.3.15** (*Association of diagrams with multigraphs*) Figure 4.4 illustrates the association of diagrams with multigraphs. Consider the diagram  $\Gamma(\pi^*, \sigma)$  in (a).  $\pi^*$  has  $p = 3$  blocks denoted  $A, B, C$  and  $\sigma$  has  $q = 3$  blocks denoted  $a, b, c$ . Hence the diagram has  $q = 3$  segments. The labeling  $A, B, C, a, b, c$  of the diagram in (a) is a multigraph labeling. We included it to make the correspondence between diagram and multigraph

apparent. The diagram in (a) is associated with the multigraph  $A \in \mathcal{A}(3, 1, 2)$  in (c) which has 3 points  $A, B, C$  of degrees 3, 1, 2 respectively and  $q = 3$  lines  $a, b, c$ . These three lines are associated with the  $q = 3$  segments of the diagram  $\Gamma(\pi^*, \sigma)$ .

Figure 4.5 displays the diagram of Figure 4.4 in (a) and also a second diagram in (b). Both diagrams are associated with the same multigraph (c).

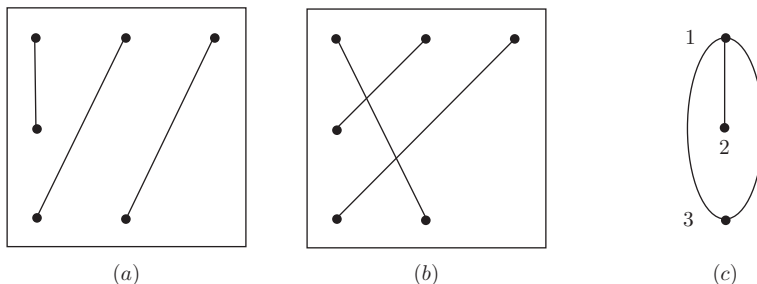
We defined connected diagrams in Definition 4.3.8. We need a related notion for multigraphs.

**Definition 4.3.16** We will say that a multigraph  $A \in \mathcal{A}_p$  is *connected* if its  $p$  points cannot be separated into two nonempty, disjoint subsets without any line connecting the points of the two subsets.

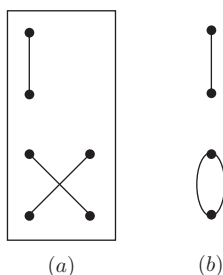
Note that connected diagrams  $\Gamma(\pi^*, \sigma)$ ,  $\sigma \in \mathcal{M}_2([n], \pi^*)$ , are associated with connected multigraphs.

**Example 4.3.17** ((Non-)Connected diagrams and multigraphs) The non-flat Gaussian diagram in Figure 4.6, (a), is not connected nor is the associated multigraph in Figure 4.6, (b).

Denote by  $\mathcal{A}_c(n_1, \dots, n_p)$  ( $\mathcal{A}_{p,c}(k)$ , resp.) the set of all connected multigraphs  $A \in \mathcal{A}(n_1, \dots, n_p)$  ( $A \in \mathcal{A}_p(k)$ , resp.).



**Figure 4.5** Non-flat, Gaussian diagrams (a) and (b), and the multigraph (c) associated with them.



**Figure 4.6** (a) A non-connected non-flat Gaussian diagram. (b) The associated non-connected multigraph.

Note that, as in Figure 4.5, different diagrams with partitions  $\sigma \in \mathcal{M}_2^0([n], \pi^*)$  can be associated with the same multigraph  $A \in \mathcal{A}(n_1, \dots, n_p)$ . The following auxiliary result provides the number of such  $\sigma$ s.

**Lemma 4.3.18** *The number of different<sup>7</sup>  $\sigma \in \mathcal{M}_2^0([n], \pi^*)$  ( $\mathcal{M}_2([n], \pi^*)$ , resp.) associated with the same multigraph  $A \in \mathcal{A}(n_1, \dots, n_p)$  ( $\mathcal{A}_c(n_1, \dots, n_p)$ , resp.) in the sense of Definition 4.3.14 is equal to*

$$\frac{n_1! \dots n_p!}{v_1! \dots v_{\binom{p}{2}}!}, \quad \text{or equivalently} \quad n_1! \dots n_p! g(A), \quad (4.3.12)$$

where  $g(A)$  is defined in (4.3.11).

*Proof* The formula (4.3.12) can be proved by induction in  $m$ , the number of pairs of points in a multigraph which have at least one connecting line. The case  $m = 1$  of smallest  $m$  occurs only when the multigraph has  $p = 2$  points with degrees  $n_1 = n_2 =: n_0$ . In this case,  $v_1 = n_0$ . The number of diagrams leading to the same multigraph is  $n_0!$ . Since

$$n_0! = \frac{n_1! n_2!}{n_0!},$$

we have (4.3.12). Suppose now that the formula (4.3.12) holds up to  $m - 1$ , and consider the general case of  $m$  above. Suppose without loss of generality that  $v_1 \geq 1$  lines connect points  $i$  and  $j$  in a multigraph. This means that  $v_1$  points in the block  $\{n_1 + \dots + n_{i-1} + 1, \dots, n_1 + \dots + n_i\}$  connect  $v_1$  points in the block  $\{n_1 + \dots + n_{j-1} + 1, \dots, n_1 + \dots + n_j\}$ . The number of such possibilities is

$$\binom{n_i}{v_1} \binom{n_j}{v_1} v_1! = \frac{n_i! n_j!}{(n_i - v_1)! (n_j - v_1)! v_1!}. \quad (4.3.13)$$

By the induction hypothesis, the number of possibilities to connect the rest of the points is

$$\frac{n_1! \dots n_{i-1}! (n_i - v_1)! n_{i+1}! \dots n_{j-1}! (n_j - v_1)! n_{j+1}! \dots n_p!}{v_2! \dots v_{\binom{p}{2}}!}, \quad (4.3.14)$$

since  $v_1$  points were already connected in the blocks  $\{n_1 + \dots + n_{i-1} + 1, \dots, n_1 + \dots + n_i\}$  and  $\{n_1 + \dots + n_{j-1} + 1, \dots, n_1 + \dots + n_j\}$ . Multiplying (4.3.13) and (4.3.14) leads to the formula (4.3.12). Note that the argument above does not depend on whether a multigraph is connected or not.  $\square$

The following is an immediate consequence of Lemma 4.3.18.

**Corollary 4.3.19** *Let  $F : \mathcal{M}_2^0([n], \pi^*) \mapsto \mathbb{R}$  be such that*

$$F(\sigma) = G(A), \quad (4.3.15)$$

<sup>7</sup> More precisely: “The number of different diagrams  $\Gamma(\pi^*, \sigma)$  with  $\sigma \in \mathcal{M}_2^0([n], \pi^*)$  associated with the same multigraph  $A \dots$ ”

where  $A \in \mathcal{A}(n_1, \dots, n_p)$  is the multigraph associated with  $\sigma \in \mathcal{M}_2^0([n], \pi^*)$ . Then,

$$\sum_{\sigma \in \mathcal{M}_2^0([n], \pi^*)} F(\sigma) = n_1! \dots n_p! \sum_{A \in \mathcal{A}(n_1, \dots, n_p)} g(A) G(A). \quad (4.3.16)$$

This result continues to hold if throughout,  $\mathcal{M}_2^0$  is replaced by  $\mathcal{M}_2$  and  $\mathcal{A}$  by  $\mathcal{A}_c$ . In particular, if  $F : \mathcal{M}_2([n], \pi^*) \mapsto \mathbb{R}$ , then

$$\sum_{\sigma \in \mathcal{M}_2([n], \pi^*)} F(\sigma) = n_1! \dots n_p! \sum_{A \in \mathcal{A}_c(n_1, \dots, n_p)} g(A) G(A). \quad (4.3.17)$$

Applying Corollary 4.3.19 to the right-hand sides of (4.3.3) and (4.3.9), leads to the following alternative diagram formulae for the moments and cumulants of multiple integrals.

**Proposition 4.3.20** For symmetric  $f_i \in L^2(E^{n_i}, m^{n_i})$ ,  $i = 1, \dots, p$ , and non-atomic  $m$ ,

$$\mathbb{E}(I_{n_1}(f_1) \dots I_{n_p}(f_p)) = n_1! \dots n_p! \sum_{A \in \mathcal{A}(n_1, \dots, n_p)} g(A) \int_{E^{n/2}} f_{A,p} dm^{n/2} \quad (4.3.18)$$

and

$$\chi(I_{n_1}(f_1), \dots, I_{n_p}(f_p)) = n_1! \dots n_p! \sum_{A \in \mathcal{A}_c(n_1, \dots, n_p)} g(A) \int_{E^{n/2}} f_{A,p} dm^{n/2}. \quad (4.3.19)$$

Each multigraph  $A$  has  $p$  points,  $q = n/2$  lines and multiplicity  $g(A)$ , defined in (4.3.11). Denote by  $\{(i_1, j_1), \dots, (i_{n/2}, j_{n/2})\}$  the  $n/2$  lines (pair sequence) of the multigraph  $A$ . The function  $f_{A,p}$  of  $n/2$  variables equals

$$f_1(u_1, \dots, u_{n_1}) f_2(u_{n_1+1}, \dots, u_{n_2}) \dots f_p(u_{n_1+\dots+n_{p-1}+1}, \dots, u_n),$$

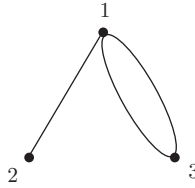
where, for each line  $(i_m, j_m)$ ,  $m = 1, \dots, n/2$ , one identifies any variable of  $f_{i_m}$  with any variable of  $f_{j_m}$ .

If  $\mathcal{A}(n_1, \dots, n_p) = \emptyset$  or  $\mathcal{A}_c(n_1, \dots, n_p) = \emptyset$  (in particular, if  $n$  is odd), the right-hand side of (4.3.18) or (4.3.19) is interpreted as 0.

*Proof* Note that, for a given  $(i_m, j_m)$ , it does not matter which two variables of  $f_{i_m}$  and  $f_{j_m}$  are identified because  $f_i$ s are symmetric. Note also that, up to a possible permutation of its  $n/2$  variables,  $f_{A,p}$  does not depend on the chosen pair sequence, that is, on the way the pair sequence is enumerated. This is because any two pair sequences describing  $A$  are the same up to a possible permutation of pairs, or indices within pairs. In particular, since  $dm^{n/2}$  is symmetric in its variables, the value of  $\int_{E^{n/2}} f_{A,p} dm^{n/2}$  does not depend on the chosen pair sequence.

The relation (4.3.18), for example, follows from (4.3.3) and (4.3.16) after observing the following. Any variables  $u_i$  and  $u_j$  identified for  $f_{\sigma,p}$  in

$$f_1(u_1, \dots, u_{n_1}) f_2(u_{n_1+1}, \dots, u_{n_2}) \dots f_p(u_{n_1+\dots+n_{p-1}+1}, \dots, u_n)$$



**Figure 4.7** The single multigraph in  $\mathcal{A}(3, 1, 2)$ .

when  $i \sim_{\sigma} j$ , correspond to a pair  $(i, j)$  in the pair sequence describing  $A$ . Then,  $f_{\sigma,p}$  is one possible representation for  $f_{A,p}$ , and

$$\int_{E^{n/2}} f_{\sigma,p} dm^{n/2} = \int_{E^{n/2}} f_{A,p} dm^{n/2},$$

so that the condition (4.3.15) holds.  $\square$

**Example 4.3.21** (*Diagram formula through multigraphs*) As in Example 4.3.5, let  $f_1$ ,  $f_2$  and  $f_3$  be respective functions of 3, 1 and 2 variables. Here  $n = 3 + 1 + 2 = 6$ . We evaluated  $\mathbb{E}I_3(f_1)I_1(f_2)I_2(f_3)$  in that example using diagrams. We will see that it can be evaluated simply by using multigraphs and applying Proposition 4.3.20. The multigraphs in  $\mathcal{A}(3, 1, 2)$  have  $p = 3$  points of degree  $n_1 = 3$ ,  $n_2 = 1$ ,  $n_3 = 2$  respectively and have  $q = n/2 = (3 + 1 + 2)/2 = 3$  lines. In fact,  $\mathcal{A}(3, 1, 2)$  contains a single multigraph  $A$  depicted in Figure 4.7 with  $g(A) = 1/2!$ . This multigraph is associated with any of the six diagrams in Example 4.3.5. A pair sequence  $\{(1, 3), (1, 3), (1, 2)\}$  describes  $A$ , and  $f_{A,p}(v_1, v_2, v_3) = f_1(v_1, v_2, v_3)f_2(v_3)f_3(v_1, v_2)$  as defined in Proposition 4.3.20. Applying (4.3.18) now yields

$$\mathbb{E}I_3(f_1)I_1(f_2)I_2(f_3) = 6 \int_{E^3} f_1(v_1, v_2, v_3)f_2(v_3)f_3(v_1, v_2)m(dv_1)m(dv_2)m(dv_3),$$

since  $n_1!n_2!n_3!/g(A) = 3!1!2!/2! = 6$ .

#### 4.3.4 Diagram and Multigraph Formulae for Hermite Polynomials

By using the connection between Hermite polynomials of Gaussian variables and multiple integrals in Proposition 4.1.2, one can obtain a diagram formula for the Hermite polynomials of Gaussian variables.

**Proposition 4.3.22** (*Diagram formula.*) Let  $(X_1, \dots, X_p)'$ ,  $p \geq 2$ , be a Gaussian vector with  $\mathbb{E}X_j = 0$ ,  $\mathbb{E}X_j^2 = 1$ ,  $j = 1, \dots, p$ . Then, for any  $n_1, \dots, n_p \geq 1$ ,

$$\mathbb{E}\left(H_{n_1}(X_1) \dots H_{n_p}(X_p)\right) = \sum_{\sigma \in \mathcal{M}_2^0([n], \pi^*)} \prod_{1 \leq i < j \leq p} (\mathbb{E}X_i X_j)^{\ell_{ij}(\sigma)}, \quad (4.3.20)$$

where  $\ell_{ij}(\sigma)$  is the number of connecting segments<sup>8</sup> between the blocks  $i$  and  $j$  of the partition  $\pi^*$  in the diagram  $\Gamma(\pi^*, \sigma)$ .

<sup>8</sup> The “segments” are the “closed curves” in Definition 4.3.1.

*Proof* We argue first that there are  $f_j \in L^2(\mathbb{R}, du)$ ,  $j = 1, \dots, p$ , such that

$$(X_1, \dots, X_p)' \stackrel{d}{=} (I_1(f_1), \dots, I_1(f_p))', \quad (4.3.21)$$

where  $I_1(f) = \int_{\mathbb{R}} f(u)B(du)$  and  $B(du)$  is a real-valued Gaussian random measure on  $\mathbb{R}$  with control measure  $du$ . By factorizing the covariance matrix of  $(X_1, \dots, X_p)$ , the following representation holds:

$$(X_1, \dots, X_p)' \stackrel{d}{=} (\phi_{11}\epsilon_1 + \dots + \phi_{1p}\epsilon_p, \dots, \phi_{p1}\epsilon_1 + \dots + \phi_{pp}\epsilon_p)', \quad (4.3.22)$$

where  $\epsilon_j$ ,  $j = 1, \dots, p$ , are i.i.d.  $\mathcal{N}(0, 1)$  random variables. Choosing a collection of orthonormal functions  $g_1, \dots, g_p$  in  $L^2(\mathbb{R}, du)$ , we can represent

$$(\epsilon_1, \dots, \epsilon_p)' \stackrel{d}{=} (I_1(g_1), \dots, I_1(g_p))'. \quad (4.3.23)$$

Substituting (4.3.23) into (4.3.22) leads to (4.3.21) with  $f_j = \sum_{m=1}^p \phi_{jm}g_m$ ,  $j = 1, \dots, m$ .

Having the representation (4.3.21), we can now apply Propositions 4.1.2 and 4.3.4 to write

$$\begin{aligned} \mathbb{E}(H_{n_1}(X_1) \dots H_{n_p}(X_p)) &= \mathbb{E}(H_{n_1}(I_1(f_1)) \dots H_{n_p}(I_1(f_p))) \\ &= \mathbb{E}(I_{n_1}(f_1^{\otimes n_1}) \dots I_{n_p}(f_p^{\otimes n_p})) = \sum_{\sigma \in \mathcal{M}_2^0([n], \pi^*)} \int_{\mathbb{R}^{n/2}} f_{\sigma,p} dm^{n/2}, \end{aligned}$$

where  $n = n_1 + \dots + n_p$  and  $m$  is the Lebesgue measure. The function  $f_{\sigma,p}$ , defined in Proposition 4.3.4, is obtained by identifying  $u_i$  and  $u_j$  in the argument of

$$f_1(u_1) \dots f_1(u_{n_1}) f_2(u_{n_1+1}) \dots f_2(u_{n_2}) \dots f_p(u_{n_1+\dots+n_{p-1}+1}) \dots f_p(u_n)$$

if and only if  $i \sim_{\sigma} j$ . The argument  $u_i$  is one of the arguments of the function  $f_i$  and the argument  $u_j$  is one of the arguments of the function  $f_j$ . It follows that

$$\int_{\mathbb{R}^{n/2}} f_{\sigma,p} dm^{n/2} = \prod_{1 \leq i < j \leq p} \left( \int_{\mathbb{R}} f_i(u) f_j(u) du \right)^{\ell_{ij}(\sigma)},$$

where  $\ell_{ij}(\sigma)$  is as in (4.3.20). Since  $\int_{\mathbb{R}} f_i(u) f_j(u) du = \mathbb{E}X_i X_j$ , we deduce (4.3.20).  $\square$

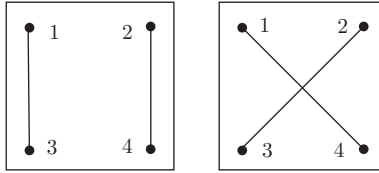
**Example 4.3.23** (*Diagram formula for Hermite polynomials*) Consider  $\mathbb{E}H_2(X_1)H_2(X_2)$ . Here,  $\pi^* = \{\{1, 2\}, \{3, 4\}\}$  and thus  $\mathcal{M}_2^0([4], \pi^*)$  contains two partitions  $\sigma$ , namely,  $\sigma_1 = \{\{1, 3\}, \{2, 4\}\}$  and  $\sigma_2 = \{\{1, 4\}, \{2, 3\}\}$ . Each one of these two partitions indicates the presence of two segments connecting the blocks  $\{1, 2\}$  and  $\{3, 4\}$  of  $\pi^*$  (see Figure 4.8). Therefore,

$$\mathbb{E}H_2(X_1)H_2(X_2) = (\mathbb{E}X_1 X_2)^{\ell_{12}(\sigma_1)} + (\mathbb{E}X_1 X_2)^{\ell_{12}(\sigma_2)} = 2(\mathbb{E}X_1 X_2)^2.$$

Here is a second perspective which also illustrates the proof of Proposition 4.3.22. If  $X_1 = I_1(f_1)$  and  $X_2 = I_1(f_2)$ , then

$$\mathbb{E}H_2(X_1)H_2(X_2) = \mathbb{E}I_2(f_1 \otimes f_1)I_2(f_2 \otimes f_2).$$





**Figure 4.8** Two diagrams in Example 4.3.23.

In  $f_1(u_1)f_1(u_2)f_2(u_3)f_2(u_4)$ , the argument  $u_1$  can be identified with  $u_3$ ,  $u_2$  with  $u_4$  but also  $u_1$  can be identified with  $u_4$  and  $u_2$  with  $u_3$ . One therefore obtains

$$2\left(\int_{\mathbb{R}} f_1(u)f_2(u)m(du)\right)^2 = 2(\mathbb{E}X_1X_2)^2.$$

For the cumulants  $\chi(H_{n_1}(X_1), \dots, H_{n_p}(X_p))$  of Hermite polynomials of Gaussian variables, the formula (4.3.20) holds where  $\mathcal{M}_2^0([n], \pi^*)$  is replaced by  $\mathcal{M}_2([n], \pi^*)$ .

**Proposition 4.3.24** (Diagram formula.) *Let  $(X_1, \dots, X_p)'$ ,  $p \geq 2$ , be a Gaussian vector with  $\mathbb{E}X_j = 0$ ,  $\mathbb{E}X_j^2 = 1$ ,  $j = 1, \dots, p$ . Then, for any  $n_1, \dots, n_p \geq 1$ ,*

$$\chi(H_{n_1}(X_1), \dots, H_{n_p}(X_p)) = \sum_{\sigma \in \mathcal{M}_2([n], \pi^*)} \prod_{1 \leq i < j \leq p} (\mathbb{E}X_iX_j)^{\ell_{ij}(\sigma)}, \quad (4.3.24)$$

where  $\ell_{ij}(\sigma)$  is the number of connecting segments between the blocks  $i$  and  $j$  of the partition  $\pi^*$  in the diagram  $\Gamma(\pi^*, \sigma)$ .

*Proof* The proof of this fact is analogous to that of Proposition 4.3.22 by using Proposition 4.3.10.  $\square$

The formulae (4.3.20) and (4.3.24) can be rewritten using *multigraphs*. We shall use the following additional notation. Denote by  $E(A)$  the set of lines (edges) in a multigraph  $A$ . For  $w \in E(A)$ , set  $d_1(w)$  and  $d_2(w)$ , with  $d_1(w) < d_2(w)$ , the two points connected by  $w$ . For a pair of points in a multigraph  $A \in \mathcal{A}_p$  indexed by  $u = 1, 2, \dots, \binom{p}{2}$ , we also denote  $u(1)$  and  $u(2)$ , with  $u(1) < u(2)$ , the two points in the pair.

**Proposition 4.3.25** *Let  $(X_1, \dots, X_p)'$ ,  $p \geq 2$ , be a Gaussian vector with  $\mathbb{E}X_j = 0$ ,  $\mathbb{E}X_j^2 = 1$ ,  $j = 1, \dots, p$ . Then, for any  $n_1, \dots, n_p \geq 1$ ,*

$$\mathbb{E}(H_{n_1}(X_1) \dots H_{n_p}(X_p)) = n_1! \dots n_p! \sum_{A \in \mathcal{A}(n_1, \dots, n_p)} g(A) \prod_{w \in E(A)} \mathbb{E}X_{d_1(w)}X_{d_2(w)} \quad (4.3.25)$$

and

$$\chi(H_{n_1}(X_1), \dots, H_{n_p}(X_p)) = n_1! \dots n_p! \sum_{A \in \mathcal{A}_c(n_1, \dots, n_p)} g(A) \prod_{w \in E(A)} \mathbb{E}X_{d_1(w)}X_{d_2(w)}, \quad (4.3.26)$$

where  $g(A)$  is defined in (4.3.11). Moreover,

$$\prod_{w \in E(A)} \mathbb{E} X_{d_1(w)} X_{d_2(w)} = \prod_{u=1}^{\binom{p}{2}} (\mathbb{E} X_{u(1)} X_{u(2)})^{v_u},$$

where  $v_u$  appears in (4.3.11).

*Proof* The relations (4.3.25) and (4.3.26) follow from (4.3.20) and (4.3.24) by applying Corollary 4.3.19.  $\square$

#### 4.4 Moments and Cumulants of Hermite Processes

Let  $\{Z_H^{(k)}(t)\}_{t \in \mathbb{R}}$  be the Hermite process of order  $k$  in Definition 4.2.1, and

$$\mu_p(t_1, \dots, t_p) = \mathbb{E} Z_H^{(k)}(t_1) \dots Z_H^{(k)}(t_p), \quad p \geq 1, t_1, \dots, t_p \in \mathbb{R}, \quad (4.4.1)$$

be its finite-dimensional moments. In the next result, we provide a useful formula for the finite-dimensional moments  $\mu_p(t_1, \dots, t_p)$ .

**Proposition 4.4.1** *Let  $\mu_p(t_1, \dots, t_p)$ ,  $p \geq 1$ ,  $t_1, \dots, t_p \in \mathbb{R}$ , be the finite-dimensional moments of the Hermite process of order  $k$  in (4.4.1). Then,*

$$\mu_p(t_1, \dots, t_p) = C_{p,k,H_0} \sum_{A \in \mathcal{A}_p(k)} g(A) S_D(A; t_1, \dots, t_p), \quad (4.4.2)$$

where  $g(A)$  is a multiplicity number of a multigraph  $A$  defined in (4.3.11), the constant is

$$C_{p,k,H_0} = (a_{k,H_0})^p \left( B(H_0 - \frac{1}{2}, 2 - 2H_0) \right)^q (k!)^p = \left( k!(1 - k(1 - H_0))(1 - 2k(1 - H_0)) \right)^{p/2} \quad (4.4.3)$$

with  $q = pk/2$  and

$$S_D(A; t_1, \dots, t_p) = \int_0^{t_1} ds_1 \dots \int_0^{t_p} ds_p |s_{i_1} - s_{j_1}|^{2H_0-2} \dots |s_{i_q} - s_{j_q}|^{2H_0-2}, \quad (4.4.4)$$

where  $(i_1, j_1), \dots, (i_q, j_q)$  is a pair sequence of  $A$ .

*Proof* Write

$$Z_H^{(k)}(t) = a_{k,H_0} I_k(f_t),$$

where the kernel function  $f_t$  is defined in (4.2.5) with (4.2.6) and  $I_k$  denotes a multiple integral. We shall use the diagram formula (4.3.3) for the moments of multiple integrals. We want to compute

$$\mu_p(t_1, \dots, t_p) = (a_{k,H_0})^p \mathbb{E} I_k(f_{t_1}) \dots I_k(f_{t_p}). \quad (4.4.5)$$

Recall from Section 4.3 that  $[pk] = \{1, 2, \dots, pk\}$ ,  $\pi^* = \{\{1, \dots, k\}, \{k+1, \dots, 2k\}, \dots, \{(p-1)k+1, \dots, pk\}\}$  is the special partition of the set  $[pk]$ , and  $\mathcal{M}_2^0([pk], \pi^*)$

denotes all partitions  $\sigma$  of  $[pk]$  so that the diagram  $\Gamma(\pi^*, \sigma)$  is non-flat and Gaussian. Then, applying the diagram formula (4.3.3),

$$\mu_p(t_1, \dots, t_p) = (a_{k, H_0})^p \sum_{\sigma \in \mathcal{M}_2^0([pk], \pi^*)} \int_{\mathbb{R}^{pk/2}} f_{\sigma, p} d(\text{Leb})^{pk/2},$$

where  $(\text{Leb})^{pk/2}$  denotes the Lebesgue measure on  $\mathbb{R}^{pk/2}$ , and  $f_{\sigma, p}$  is a function of  $pk/2$  variables obtained by identifying the variables  $u_i$  and  $u_j$  in

$$f_{t_1}(u_1, \dots, u_k) f_{t_2}(u_{k+1}, \dots, u_{2k}) \dots f_{t_p}(u_{(p-1)k+1}, \dots, u_{pk})$$

if and only if  $i \sim_{\sigma} j$ . In view of (4.2.5), we can also write<sup>9</sup>

$$\mu_p(t_1, \dots, t_p) = (a_{k, H_0})^p \sum_{\sigma \in \mathcal{M}_2^0([pk], \pi^*)} \int_0^{t_1} ds_1 \dots \int_0^{t_p} ds_p \left\{ \int_{\mathbb{R}^{pk/2}} g_{\sigma, p} d(\text{Leb})^{pk/2} \right\}, \quad (4.4.6)$$

where  $g_{\sigma, p}$  is obtained by identifying the variables  $u_i$  and  $u_j$  in

$$g_{s_1}(u_1, \dots, u_k) g_{s_2}(u_{k+1}, \dots, u_{2k}) \dots g_{s_p}(u_{(p-1)k+1}, \dots, u_{pk}) \quad (4.4.7)$$

and where

$$g_s(u_1, \dots, u_k) = \prod_{j=1}^k (s - u_j)_+^{H_0 - \frac{3}{2}}.$$

If  $u_i = u_j = u$  are identified in (4.4.7), and  $u_i$  enters in  $g_{s_i}$  and  $u_j$  enters in  $g_{s_j}$ , then the integral  $\int_{\mathbb{R}^{pk/2}} g_{\sigma, p} d(\text{Leb})^{pk/2}$  in (4.4.6) over  $u$  yields the integral  $\int_{\mathbb{R}} (s_i - u)_+^{H_0 - \frac{3}{2}} (s_j - u)_+^{H_0 - \frac{3}{2}} du$ . For example, if  $s_j > s_i$ , by making the change of variables  $s_i - u = (s_j - s_i)v$ , we have

$$\int_{\mathbb{R}} (s_i - u)_+^{H_0 - \frac{3}{2}} (s_j - u)_+^{H_0 - \frac{3}{2}} du = \int_0^{\infty} v_+^{H_0 - \frac{3}{2}} (1 + v)_+^{H_0 - \frac{3}{2}} dv (s_j - s_i)^{2H_0 - 2}.$$

Since the integral is symmetric in  $s_i$  and  $s_j$  and in view of (2.2.12), we conclude that

$$\int_{\mathbb{R}} (s_i - u)_+^{H_0 - \frac{3}{2}} (s_j - u)_+^{H_0 - \frac{3}{2}} du = B(H_0 - 1/2, 2 - 2H_0) |s_i - s_j|^{2H_0 - 2}. \quad (4.4.8)$$

Observe that the function in (4.4.7) has  $pk$  arguments and hence  $q = pk/2$  pairs of identified arguments. Thus, if  $(i_1, j_1), \dots, (i_q, j_q)$ ,  $q = pk/2$ , denote all the indices  $(i, j)$  of  $s_i$  and  $s_j$  associated with these pairs, we get

$$\int_{\mathbb{R}^{pk/2}} g_{\sigma, p} d(\text{Leb})^{pk/2} = (B(H_0 - 1/2, 2 - 2H_0))^q |s_{i_1} - s_{j_1}|^{2H_0 - 2} \dots |s_{i_q} - s_{j_q}|^{2H_0 - 2}.$$

Note that  $i_m \neq j_m$ ,  $m = 1, \dots, q$ , since  $\Gamma(\pi^*, \sigma)$  is non-flat. Moreover, every index  $1, \dots, p$  appears exactly  $k$  times in the sequence  $(i_1, j_1), \dots, (i_q, j_q)$ , since  $\Gamma(\pi^*, \sigma)$  is

<sup>9</sup> Do not confuse  $p = H_0 - 1/2$  in (4.2.5) and (4.2.6) with  $p$  in (4.4.1).

Gaussian.<sup>10</sup> The sequence  $(i_1, j_1), \dots, (i_q, j_q)$  can therefore be thought as a pair sequence of a multigraph  $A \in \mathcal{A}_p(k)$ . Then,

$$\int_0^{t_1} ds_1 \dots \int_0^{t_p} ds_p \left\{ \int_{\mathbb{R}^{pk/2}} g_{\sigma,p} d(\text{Leb})^{pk/2} \right\} = (B(H_0 - 1/2, 2 - 2H_0))^q S_D(A; t_1, \dots, t_p), \quad (4.4.9)$$

where  $S_D(A; t_1, \dots, t_p)$  is defined in (4.4.4).

Every  $\sigma \in \mathcal{M}_2^0([pk], \pi^*)$  is thus associated with a multigraph  $A \in \mathcal{A}_p(k)$ , and the relation (4.4.9) holds. However, a number of different partitions  $\sigma \in \mathcal{M}_2^0([pk], \pi^*)$  lead to the same multigraph. By Lemma 4.3.18, that number is given by

$$\frac{(k!)^p}{v_1! \dots v_{\binom{p}{2}}!} = (k!)^p g(A), \quad (4.4.10)$$

where  $g(A)$  appears in (4.3.11). The numerator in (4.4.10) is  $(k!)^p$  because each one of the  $p$  points of a multigraph  $A \in \mathcal{A}_p(k)$  has degree  $k$ . Finally, the result (4.4.2) follows from (4.4.6), (4.4.9) and (4.4.10).  $\square$

**Remark 4.4.2** One can show (Exercise 4.4) that

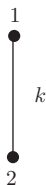
$$\mu_p(t_1, \dots, t_p) = \frac{C_{p,k,H_0}}{2^q (q!)^1} \sum_1 \int_0^{t_1} ds_1 \dots \int_0^{t_p} ds_p |s_{i_1} - s_{j_1}|^{2H_0-2} \dots |s_{i_q} - s_{j_q}|^{2H_0-2}, \quad (4.4.11)$$

where  $q = pk/2$  and the sum  $\sum_1$  is over all indices  $i_1, j_1, \dots, i_q, j_q$  such that

- (i)  $i_1, j_1, \dots, i_q, j_q \in \{1, 2, \dots, p\}$ ,
- (ii)  $i_1 \neq j_1, \dots, i_q \neq j_q$ ,
- (iii) each number  $1, 2, \dots, p$  appears exactly  $k$  times in  $(i_1, j_1, \dots, i_q, j_q)$ .

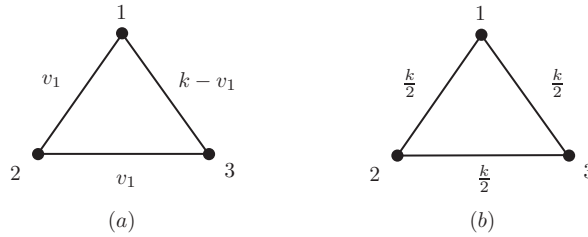
It is interesting to examine the formula (4.4.2) in the special cases of the first few values  $p = 2, 3$  and  $4$ .

When  $p = 2$  and  $k \geq 1$ , there is only one multigraph  $A$  in  $\mathcal{A}_2(k)$  with  $k$  lines connecting points 1 and 2, and with  $g(A) = 1/k!$ . It is depicted in Figure 4.9. Then, for  $t_1, t_2 \geq 0$ , the formula (4.4.2) becomes



**Figure 4.9** Multigraphs in  $\mathcal{A}_3(k)$ . The number of lines between any two points in a multigraph of the set  $\mathcal{A}_2(k)$  is indicated by a number next to the line connecting the two points.

<sup>10</sup> For example: the index 2 appears exactly  $k = 3$  times in the sequence  $\{(1, 2), (2, 3), (3, 4), (4, 1), (1, 3), (2, 4)\}$ .



**Figure 4.10** The number of lines between any two points in a multigraph of the set is indicated by a number next to the line connecting the two points.

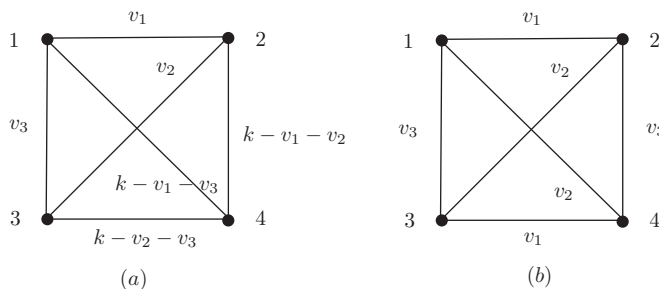
$$\begin{aligned}
 \mu_2(t_1, t_2) &= \frac{C_{2,k,H_0}}{k!} \int_0^{t_1} ds_1 \int_0^{t_2} ds_2 |s_1 - s_2|^{k(2H_0-2)} \\
 &= \frac{C_{2,k,H_0}}{k!(1-2k(1-H_0))(2-2k(1-H_0))} \left\{ t_1^{2-2k(1-H_0)} + t_2^{2-2k(1-H_0)} - |t_1 - t_2|^{2-2k(1-H_0)} \right\} \\
 &= \frac{1}{2} \left\{ t_1^{2H} + t_2^{2H} - |t_1 - t_2|^{2H} \right\} \quad (4.4.12)
 \end{aligned}$$

by using (4.4.3) and (4.2.1). We thus get the covariance function of an  $H$ -self-similar process with stationary increments (Proposition 2.5.6, (e)).

When  $p = 3$ , let  $v_1$  be the number of lines between points 1 and 2 in a multigraph  $A \in \mathcal{A}_3(k)$ . Recall that every multigraph in  $\mathcal{A}_3(k)$  has three points (vertices) and each point has degree  $k$ . For point 1 to have degree  $k$ , the number of lines between 1 and 3 must be  $k - v_1$  and then, for point 3 to have degree  $k$ , the number of lines between the points 2 and 3 must be  $v_1$  (Figure 4.10, (a)). In order that point 2 should have degree  $k$  as well, we must have  $2v_1 = k$  and hence  $v_1 = k/2$  which is an integer only when  $k$  is even. Thus, when  $k$  is even, there is only one multigraph  $A \in \mathcal{A}_3(k)$  depicted in Figure 4.10, (b), and with  $g(A) = 1/(k/2!)^3$ . The formula (4.4.2) becomes: when  $k$  is even,

$$\begin{aligned}
 \mu_3(t_1, t_2, t_3) &= \frac{C_{3,k,H_0}}{((k/2)!)^3} \int_0^{t_1} ds_1 \int_0^{t_2} ds_2 \int_0^{t_3} ds_3 |s_1 - s_2|^{\frac{k}{2}(2H_0-2)} |s_2 - s_3|^{\frac{k}{2}(2H_0-2)} |s_3 - s_1|^{\frac{k}{2}(2H_0-2)} \\
 &= \frac{C_{3,k,H_0}}{((k/2)!)^3} \int_0^{t_1} ds_1 \int_0^{t_2} ds_2 \int_0^{t_3} ds_3 |s_1 - s_2|^{k(H_0-1)} |s_2 - s_3|^{k(H_0-1)} |s_3 - s_1|^{k(H_0-1)}. \quad (4.4.13)
 \end{aligned}$$

Suppose now  $p = 4$  and  $k \geq 1$ . Let  $v_1, v_2$  and  $v_3$  be the number of lines in a multigraph  $A \in \mathcal{A}_4(k)$  between points 1 and 2, 2 and 3, and 1 and 3, respectively. Then, since each point has degree  $k$ , the number of lines between points 1 and 4, 2 and 4, and 2 and 3 are  $k - v_1 - v_3, k - v_1 - v_2$  and  $k - v_2 - v_3$ , respectively. See Figure 4.11, (a). Since point 4 has degree  $k$ ,  $(k - v_1 - v_3) + (k - v_1 - v_2) + (k - v_2 - v_3) = k$  or  $v_1 + v_2 + v_3 = k$ . This implies that the number of lines between points 1 and 4, 2 and 4, and 2 and 3 are  $v_2, v_3$  and  $v_1$ , respectively. See Figure 4.11, (b). In the case  $p = 4$ , the formula (4.4.2) can now be written as



**Figure 4.11** The number of lines between any two points in a multigraph of the set  $\mathcal{A}_4(k)$  is indicated by a number next to the line connecting the two points.

$$\begin{aligned} \mu_4(t_1, t_2, t_3, t_4) = C_{4,k,H_0} \sum_{\substack{v_1, v_2, v_3 \geq 0 \\ v_1 + v_2 + v_3 = k}} \frac{1}{(v_1! v_2! v_3!)^2} \int_0^{t_1} ds_1 \int_0^{t_2} ds_2 \int_0^{t_3} ds_3 \int_0^{t_4} ds_4 \\ \times |(s_1 - s_2)(s_3 - s_4)|^{v_1(2H_0-2)} |(s_1 - s_4)(s_2 - s_3)|^{v_2(2H_0-2)} |(s_1 - s_3)(s_2 - s_4)|^{v_3(2H_0-2)}. \end{aligned} \quad (4.4.14)$$

The formulae above concern the moments of Hermite processes. There are analogous formulae for the cumulants. Let

$$\chi_p(t_1, \dots, t_p) = \chi(Z_H^{(k)}(t_1), \dots, Z_H^{(k)}(t_p)), \quad p \geq 1, t_1, \dots, t_p \in \mathbb{R}, \quad (4.4.15)$$

where the joint cumulant  $\chi$  is defined in (4.3.7).

**Proposition 4.4.3** Let  $\chi_p(t_1, \dots, t_p)$ ,  $p \geq 1$ ,  $t_1, \dots, t_p \in \mathbb{R}$ , be the joint cumulants of the Hermite process of order  $k$  in (4.4.15). Then,

$$\chi_p(t_1, \dots, t_p) = C_{p,k,H_0} \sum_{A \in \mathcal{A}_{p,c}(k)} g(A) S_D(A; t_1, \dots, t_p), \quad (4.4.16)$$

where  $C_{p,k,H_0}$ ,  $g(A)$  and  $S_D(A; t_1, \dots, t_p)$  are as in Proposition 4.4.1.

*Proof* The result can be proved as Proposition 4.4.1 by using the formula (4.3.9), and by observing that the partitions  $\sigma \in \mathcal{M}_2([pk], \pi^*)$  entering the formula (4.3.9) are now associated with connected multigraphs  $A \in \mathcal{A}_{p,c}(k)$  (see also Proposition 4.3.25).  $\square$

When the mean is 0, the second and third cumulants are equal to the second and third moments respectively. Therefore, the formulae (4.4.12) and (4.4.13) hold as well for  $\chi_2(t_1, t_2)$  and  $\chi_3(t_1, t_2, t_3)$  respectively. As for  $\chi_4$ , all we need is to require in addition that the multigraph be connected, thus excluding the terms  $v_1 = 0$ ,  $v_2 = 0$  or  $v_3 = 0$  in (4.4.14). Therefore,

$$\begin{aligned} \chi_4(t_1, t_2, t_3, t_4) = C_{4,k,H_0} \sum_{\substack{v_1, v_2, v_3 \geq 1 \\ v_1 + v_2 + v_3 = k}} \frac{1}{(v_1! v_2! v_3!)^2} \int_0^{t_1} ds_1 \int_0^{t_2} ds_2 \int_0^{t_3} ds_3 \int_0^{t_4} ds_4 \\ \times |(s_1 - s_2)(s_3 - s_4)|^{v_1(2H_0-2)} |(s_1 - s_4)(s_2 - s_3)|^{v_2(2H_0-2)} |(s_1 - s_3)(s_2 - s_4)|^{v_3(2H_0-2)}. \end{aligned} \quad (4.4.17)$$

### 4.5 Multiple Integrals of Order Two

Consider first a general multiple integral of order two, namely,

$$I_2(f) = \int_{E^2}' f(u_1, u_2) W(du_1) W(du_2), \quad (4.5.1)$$

where  $W$  is a Gaussian random measure on  $\mathbb{R}$  with control measure  $m$  and  $f \in L^2(E^2, m^2)$ . A number of useful results are available for such integrals. The following proposition gives its  $p$ th cumulants, where  $p \geq 2$ .

**Proposition 4.5.1** *The  $p$ th cumulant of (4.5.1),  $p \geq 2$ , is*

$$\begin{aligned} \chi_p(I_2(f)) &= \chi(\underbrace{I_2(f), \dots, I_2(f)}_{p \text{ times}}) \\ &= 2^{p-1}(p-1)! \int_{E^p} f(u_1, u_2) f(u_2, u_3) \dots f(u_{p-1}, u_p) f(u_p, u_1) m(du_1) \dots m(du_p). \end{aligned} \quad (4.5.2)$$

*Proof* Consider first the case  $p = 2$ . The only multigraph in  $\mathcal{A}_{2,c}(2)$  with two points of degree 2 is the one depicted in Figure 4.9 with  $k = 2$ , for which  $g(A) = 1/2!$ . By (4.3.19),

$$\chi_2(I_2(f)) = 2!2! \frac{1}{2!} \int_{E^2} f(u_1, u_2)^2 m(du_1) m(du_2),$$

which proves (4.5.2) for  $p = 2$ .

Suppose now  $p \geq 3$ . We shall use again an argument based on multigraphs<sup>11</sup> and apply Proposition 4.3.20. Since  $n_1 = n_2 = \dots = n_p = 2$  and  $n = n_1 + \dots + n_p = 2p$ , we have

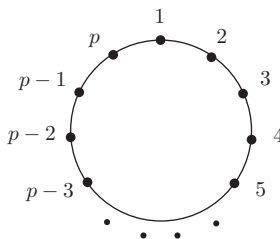
$$\chi_p(I_2(f)) = (2!)^p \sum_{A \in \mathcal{A}_{p,c}(2)} g(A) \int_{E^p} f_{A,p} dm^p,$$

where  $\mathcal{A}_{p,c}(2) = \mathcal{A}_c(2, \dots, 2)$  is the set of all connected multigraphs whose  $p \geq 3$  points have each degree 2. Observe that no  $A \in \mathcal{A}_{p,c}(2)$  can have multiple lines and hence  $g(A) = 1$ . One multigraph  $A \in \mathcal{A}_{p,c}(2)$  is depicted in Figure 4.12. All the other multigraphs in  $\mathcal{A}_{p,c}(2)$  involve a permutation of the points. Two such permutations correspond to the same multigraph if and only if the corresponding “circles” (as in Figure 4.12) are possible rotations and flipped images of each other. There are  $(p-1)!/2$  such permutations corresponding to different multigraphs. Here,  $(p-1)!$  counts possible permutations after fixing a point, say 1, and  $1/2$  takes into account mirror images after flipping a circle.

The multigraph in Figure 4.12 is described by a pair sequence  $\{(p, 1), (1, 2), (1, 3), \dots, (p-1, p)\}$ , for which

$$f_{A,p}(u_1, \dots, u_p) = f(u_1, u_2) f(u_2, u_3) \dots f(u_p, u_1). \quad (4.5.3)$$

<sup>11</sup> For an argument based on diagrams instead, see Peccati and Taqqu [797], p. 139



**Figure 4.12** A multigraph in  $\mathcal{A}_{p,c}(2)$ .

For any other multigraph in  $\mathcal{A}_{p,c}(2)$ ,  $f_{A,p}$  is defined by permuting the indices in (4.5.3). Hence,

$$\int_{E^p} f_{A,p} dm^p = \int_{E^p} f(u_1, u_2) f(u_2, u_3) \dots f(u_p, u_1) m(du_1) \dots m(du_p)$$

for all  $A \in \mathcal{A}_{p,c}(2)$ . This yields

$$\chi_p(I_2(f)) = (2!)^p \frac{(p-1)!}{2} \int_{E^p} f(u_1, u_2) f(u_2, u_3) \dots f(u_p, u_1) m(du_1) \dots m(du_p),$$

which is what we wanted to prove.  $\square$

For symmetric  $f \in L^2(E^2, m^2)$ , that is,  $f(u, v) = f(v, u)$  and  $\int_E \int_E f(u, v)^2 m(du) m(dv) < \infty$ ,

$$(\mathcal{I}g)(v) = \int_E g(u) f(u, v) m(du)$$

defines an operator from  $g \in L^2(E, m)$  to  $\mathcal{I}g \in L^2(E, m)$ , which is compact<sup>12</sup> and self-adjoint<sup>13</sup> (e.g., Conway [261], Example 2.9 and Proposition 4.7 in Chapter II). By the spectral theorem (e.g., Conway [261], Theorem 5.1 and Corollary 5.4 in Chapter II), the operator has real eigenvalues  $\lambda_k$  and orthonormal eigenfunctions  $e_k(u)$  satisfying

$$\lambda_k e_k(v) = \int_E e_k(u) f(u, v) m(du), \quad k \geq 1, \quad (4.5.4)$$

and one has

$$(\mathcal{I}g)(v) = \int_E g(u) f(u, v) m(du) = \sum_{k=1}^{\infty} \lambda_k \left( \int_E g(u) e_k(u) m(du) \right) e_k(v). \quad (4.5.5)$$

Though we index the eigenvalues by  $k \geq 1$ , the number of nonzero eigenvalues can be finite.<sup>14</sup> Note also that the relation (4.5.5) makes sense informally since it certainly holds

<sup>12</sup> Let  $X$  and  $Y$  be normed spaces, for example,  $L^2(E, m)$  as above. An operator  $T : X \rightarrow Y$  is called a *compact linear operator* if  $T$  is linear and if for every bounded subset  $B$  of  $X$ , the image  $T(B)$  has a compact closure  $\overline{T(B)}$ .

<sup>13</sup> Let  $T : H \rightarrow H$  be a bounded linear operator on a Hilbert space  $H$  with inner product  $(\cdot, \cdot)$ . Its *adjoint*  $T^*$  is such that  $(Tx, y) = (x, T^*y)$ ,  $x, y \in H$ .  $T$  is *self-adjoint* if  $T = T^*$ . Here,  $H = L^2(E, m)$  and  $(g_1, g_2) = \int_E g_1(u) g_2(u) m(du)$ . The operator  $\mathcal{I}$  is self-adjoint because the kernel  $f(u, v)$  is symmetric.

<sup>14</sup> As in the case  $f(u, v) = h(u)h(v)$  when  $(\mathcal{I}g)(v) = (\int_E g(u)h(u)du)h(v)$  has only one nonzero eigenvalue  $\lambda_1 = \int_E h(u)^2 du$  with the corresponding eigenvector  $e_1(v) = h(v)$ .



for  $g(u) = e_m(u)$ , with the left-hand side equal to  $\lambda_m e_m(u)$  by (4.5.4) and the right-hand side being the same by the orthonormality of the eigenfunctions  $e_k(u)$ ,  $k \geq 1$ . The following basic result is useful when working with multiple integral of order two.

**Lemma 4.5.2** For symmetric  $f \in L^2(E^2, m^2)$ , one has

$$I_2(f) = \sum_{k=1}^{\infty} \lambda_k (\xi_k^2 - 1), \quad (4.5.6)$$

where the convergence is in  $L^2(\Omega)$ ,  $\lambda_k$ ,  $k \geq 1$ , are the eigenvalues appearing in (4.5.4), and  $\xi_k = I_1(e_k)$ ,  $k \geq 1$ , with the eigenfunctions  $e_k$  appearing in (4.5.4). The random variables  $\xi_k$ ,  $k \geq 1$ , are i.i.d.  $\mathcal{N}(0, 1)$  random variables.

*Proof* Recall that  $\int_E e_k(u) e_j(u) m(du) = \delta_{k,j}$  by orthogonality in  $L^2(E, m)$ . We first show that

$$f(u, v) = \sum_{k=1}^{\infty} \lambda_k e_k(u) e_k(v) \quad (4.5.7)$$

a.e.  $m(du)m(dv)$ . Since  $f \in L^2(E^2, m^2)$ , by Fubini's theorem,  $\int_E f(u, v)^2 m(du) < \infty$  a.e.  $m(dv)$ . Taking  $g(u) = g_v(u) = f(u, v)$  in (4.5.5) and using (4.5.4) and (4.5.5) yields

$$\begin{aligned} \int_E f(u, v)^2 m(du) &= \int_E g_v(u) f(u, v) m(du) = \sum_{k=1}^{\infty} \lambda_k \left( \int_E f(u, v) e_k(u) m(du) \right) e_k(v) \\ &= \sum_{k=1}^{\infty} (\lambda_k e_k(v))^2 < \infty \quad \text{a.e. } m(dv). \end{aligned} \quad (4.5.8)$$

Since by the orthonormality of  $e_k$  in  $L^2(E, m)$ ,

$$\sum_{k=1}^{\infty} (\lambda_k e_k(v))^2 = \int_E \left( \sum_{k=1}^{\infty} (\lambda_k e_k(v)) e_k(u) \right)^2 m(du),$$

this shows that  $\sum_{k=1}^{\infty} \lambda_k e_k(u) e_k(v) = \sum_{k=1}^{\infty} (\lambda_k e_k(v)) e_k(u)$  converges in  $L^2(E, m)$ , a.e.  $m(dv)$ . Then, (4.5.5) can be written as

$$\int_E g(u) \left( f(u, v) - \sum_{k=1}^{\infty} \lambda_k e_k(u) e_k(v) \right) m(du) = 0$$

for a.e.  $m(dv)$ . Taking  $g(u) = g_v(u) = f(u, v) - \sum_{k=1}^{\infty} \lambda_k e_k(u) e_k(v)$  gives

$$\int_E \left( f(u, v) - \sum_{k=1}^{\infty} \lambda_k e_k(u) e_k(v) \right)^2 m(du) = 0$$

a.e.  $m(dv)$ , which yields (4.5.7).

Moreover, by (4.5.8),

$$\begin{aligned}\sum_{k=1}^{\infty} \lambda_k^2 &= \sum_{k=1}^{\infty} \lambda_k^2 \int_E e_k(v)^2 m(dv) = \int_E \sum_{k=1}^{\infty} (\lambda_k e_k(u))^2 m(dv) \\ &= \int_E \int_E f(u, v)^2 m(du) m(dv) < \infty,\end{aligned}$$

since  $f \in L^2(E^2, m^2)$ . Since  $e_k$  are orthonormal and  $\sum_{k=1}^{\infty} \lambda_k^2 < \infty$ , the series (4.5.7) also converges in  $L^2(E^2, m^2)$ . Since  $\mathbb{E}(I_2(f))^2 = 2 \int_E \int_E f(u, v)^2 m(du) m(dv) < \infty$ , we can then apply  $I_2$  to both sides of (4.5.7) to get

$$I_2(f) = \sum_{k=1}^{\infty} \lambda_k I_2(e_k^{\otimes 2}) = \sum_{k=1}^{\infty} \lambda_k H_2(I_1(e_k)),$$

by using Proposition 4.1.2, (a). It remains to observe that since  $e_k$  are orthonormal,  $I_1(e_k) =: \xi_k$  are i.i.d.  $\mathcal{N}(0, 1)$  random variables, and that  $H_2(\xi_k) = \xi_k^2 - 1$ .  $\square$

In particular, the characteristic function of  $I_2(f)$  has the product expansion

$$\mathbb{E} e^{i\theta I_2(f)} = \prod_{k=1}^{\infty} \mathbb{E} e^{i\theta \lambda_k (\xi_k^2 - 1)} = \prod_{k=1}^{\infty} (1 - i2\theta \lambda_k)^{-1/2} e^{-i\theta \lambda_k}, \quad (4.5.9)$$

where  $\xi_k^2$  are i.i.d. chi-square with 1 degree of freedom. This expression can be rewritten as

$$\begin{aligned}\mathbb{E} e^{i\theta I_2(f)} &= \exp \left\{ - \sum_{k=1}^{\infty} \left( \frac{1}{2} \log(1 - i2\theta \lambda_k) + i\theta \lambda_k \right) \right\} = \exp \left\{ \sum_{k=1}^{\infty} \left( \frac{1}{2} \sum_{p=1}^{\infty} \frac{(i2\theta \lambda_k)^p}{p} - i\theta \lambda_k \right) \right\} \\ &= \exp \left\{ \frac{1}{2} \sum_{k=1}^{\infty} \sum_{p=2}^{\infty} \frac{(i2\theta \lambda_k)^p}{p} \right\} = \exp \left\{ \frac{1}{2} \sum_{p=2}^{\infty} \frac{(i2\theta)^p}{p} \sum_{k=1}^{\infty} \lambda_k^p \right\}.\end{aligned}$$

By substituting (4.5.7) into (4.5.2), it can be shown easily that

$$\sum_{k=1}^{\infty} \lambda_k^p = \frac{\chi_p(I_2(f))}{2^{p-1}(p-1)!}, \quad p \geq 2 \quad (4.5.10)$$

(Exercise 4.5). Hence

$$\mathbb{E} e^{i\theta I_2(f)} = \exp \left\{ \frac{1}{2} \sum_{p=2}^{\infty} \frac{(i2\theta)^p}{p} \frac{\chi_p(I_2(f))}{2^{p-1}(p-1)!} \right\}, \quad (4.5.11)$$

by using the fact that  $\chi_1(I_2(f)) = \mathbb{E} I_2(f) = 0$ .

## 4.6 The Rosenblatt Process

We shall now apply the results of Section 4.5 to the Rosenblatt process which is the Hermite process  $Z_H^{(k)}$  with  $k = 2$  (see Definition 4.2.1). It can be found in various forms in Proposition 2.6.16 as well as in Definition 4.2.1. We present it here again for convenience.

**Definition 4.6.1** The *Rosenblatt process* is the Hermite process of order two defined by

$$Z_H^{(2)}(t) = a_{2,H_0} \int_{\mathbb{R}^2} \left\{ \int_0^t (s - u_1)_+^{H_0 - \frac{3}{2}} (s - u_2)_+^{H_0 - \frac{3}{2}} ds \right\} B(du_1) B(du_2), \quad t \in \mathbb{R},$$

where

$$\frac{3}{4} < H_0 < 1.$$

It is standard, that is,  $\mathbb{E} Z_H^{(2)}(1)^2 = 1$ , when

$$a_{2,H_0} = \frac{\sqrt{(H_0 - 1/2)(4H_0 - 3)}}{B(H_0 - 1/2, 2 - 2H_0)} = \frac{\sqrt{(H/2)(2H - 1)}}{B(H/2, 1 - H)}.$$

Here  $H = 2H_0 - 1 \in (1/2, 1)$  is the self-similarity parameter of the Rosenblatt process.

Other representations can be found in Theorem 4.2.10.

The following proposition gives the joint cumulants and characteristic function of the Rosenblatt process.

**Proposition 4.6.2** Let  $Z_H^{(2)} = \{Z_H^{(2)}(t)\}_{t \in \mathbb{R}}$  be the Hermite process of order two, that is, the Rosenblatt process. For  $\theta_1, \dots, \theta_J \in \mathbb{R}$ ,  $t_1, \dots, t_J \in \mathbb{R}$ , the  $p$ th cumulant,  $p \geq 2$ , and the characteristic function of a linear combination  $\sum_{j=1}^J \theta_j Z_H^{(2)}(t_j)$  are given by

$$\chi_p \left( \sum_{j=1}^J \theta_j Z_H^{(2)}(t_j) \right) = 2^{-1} (p-1)! C_{p,2,H_0} \sum_{j_1, \dots, j_p=1}^J \theta_{j_1} \dots \theta_{j_p} S(t_{j_1}, \dots, t_{j_p}) \quad (4.6.1)$$

and

$$\mathbb{E} \exp \left\{ i \sum_{j=1}^J \theta_j Z_H^{(2)}(t_j) \right\} = \exp \left\{ \frac{1}{2} \sum_{p=2}^{\infty} \frac{i^p C_{p,2,H_0}}{p} \sum_{j_1, \dots, j_p=1}^J \theta_{j_1} \dots \theta_{j_p} S(t_{j_1}, \dots, t_{j_p}) \right\}, \quad (4.6.2)$$

where  $C_{p,2,H_0}$  is defined in (4.4.3) and

$$\begin{aligned} & S(t_{j_1}, \dots, t_{j_p}) \\ &= \int_0^{t_{j_1}} ds_1 \dots \int_0^{t_{j_p}} ds_p |s_1 - s_2|^{2H_0-2} |s_2 - s_3|^{2H_0-2} \dots |s_{p-1} - s_p|^{2H_0-2} |s_p - s_1|^{2H_0-2}. \end{aligned} \quad (4.6.3)$$

*Proof* Write

$$\sum_{j=1}^J \theta_j Z_H^{(2)}(t_j) = a_{2,H_0} I_2 \left( \sum_{j=1}^J \theta_j f_{t_j} \right),$$

where the kernel function

$$f_t(u_1, u_2) = \int_0^t (s - u_1)_+^{H_0 - 3/2} (s - u_2)_+^{H_0 - 3/2} ds$$

is defined in (4.2.5) with (4.2.6) and  $I_2$  denotes a multiple integral of order two. By using (4.5.2),

$$\begin{aligned}
 \chi_p \left( \sum_{j=1}^J \theta_j Z_H^{(2)}(t_j) \right) &= (a_{2,H_0})^p 2^{p-1} (p-1)! \\
 &\times \int_{\mathbb{R}^p} \sum_{j_1=1}^J \theta_{j_1} f_{t_{j_1}}(u_1, u_2) \sum_{j_2=1}^J \theta_{j_2} f_{t_{j_2}}(u_2, u_3) \dots \sum_{j_p=1}^J \theta_{j_p} f_{t_{j_p}}(u_p, u_1) du_1 du_2 \dots du_p \\
 &= (a_{2,H_0})^p 2^{p-1} (p-1)! \sum_{j_1, \dots, j_p=1}^J \theta_{j_1} \dots \theta_{j_p} \int_{\mathbb{R}^p} f_{t_{j_1}}(u_1, u_2) \dots f_{t_{j_p}}(u_p, u_1) du_1 du_2 \dots du_p \\
 &= (a_{2,H_0})^p 2^{p-1} (p-1)! \sum_{j_1, \dots, j_p=1}^J \theta_{j_1} \dots \theta_{j_p} \int_{\mathbb{R}^p} du_1 \dots du_p \int_0^{t_{j_1}} ds_1 \dots \int_0^{t_{j_p}} ds_p \\
 &\times (s_1 - u_1)_+^{H_0 - \frac{3}{2}} (s_1 - u_2)_+^{H_0 - \frac{3}{2}} (s_2 - u_2)_+^{H_0 - \frac{3}{2}} (s_2 - u_3)_+^{H_0 - \frac{3}{2}} \dots (s_p - u_p)_+^{H_0 - \frac{3}{2}} (s_p - u_1)_+^{H_0 - \frac{3}{2}} \\
 &= (a_{2,H_0})^p 2^{p-1} (p-1)! \sum_{j_1, \dots, j_p=1}^J \theta_{j_1} \dots \theta_{j_p} \int_0^{t_{j_1}} ds_1 \dots \int_0^{t_{j_p}} ds_p \\
 &\times \left( \int_{\mathbb{R}} (s_1 - u_2)_+^{H_0 - \frac{3}{2}} (s_2 - u_2)_+^{H_0 - \frac{3}{2}} du_2 \right) \left( \int_{\mathbb{R}} (s_2 - u_3)_+^{H_0 - \frac{3}{2}} (s_3 - u_3)_+^{H_0 - \frac{3}{2}} du_3 \right) \times \dots \\
 &\times \left( \int_{\mathbb{R}} (s_{p-1} - u_p)_+^{H_0 - \frac{3}{2}} (s_p - u_p)_+^{H_0 - \frac{3}{2}} du_p \right) \left( \int_{\mathbb{R}} (s_1 - u_1)_+^{H_0 - \frac{3}{2}} (s_p - u_1)_+^{H_0 - \frac{3}{2}} du_1 \right) \\
 &= 2^{-1} (p-1)! (a_{2,H_0})^p 2^p (B(H_0 - 1/2, 2 - 2H_0))^p \sum_{j_1, \dots, j_p=1}^J \theta_{j_1} \dots \theta_{j_p} S(t_{j_1}, \dots, t_{j_p}),
 \end{aligned}$$

by using (4.4.8) and (4.6.3). Using the expression (4.4.3) for  $C_{p,2,H_0}$  yields the formula (4.6.1) since  $k = 2$  and  $q = pk/2 = p$ . The formula (4.6.2) is a consequence of (4.6.1) and

$$\mathbb{E} \exp \left\{ i \sum_{j=1}^J \theta_j Z_H^{(2)}(t_j) \right\} = \exp \left\{ \sum_{p=2}^{\infty} \frac{i^p}{p!} \chi_p \left( \sum_{j=1}^J \theta_j Z_H^{(2)}(t_j) \right) \right\}$$

since the mean (first cumulant) is zero (see also (4.5.11)). □

## 4.7 The Rosenblatt Distribution

We provide some results and observations on the Rosenblatt distribution, drawing from the work of Veillette and Taqqu [979]. For notational simplicity, we set

$$D = 2(1 - H_0) = 1 - H \in \left(0, \frac{1}{2}\right), \quad (4.7.1)$$

associated with (4.2.2) when  $k = 2$ .

**Definition 4.7.1** The *Rosenblatt distribution* is defined as the distribution of the random variable  $Z_H^{(2)}(1)$ , where  $Z_H^{(2)}$  is the (standard) Rosenblatt process. A random variable having the Rosenblatt distribution will be denoted as  $Z_D$ . Though for shortness sake, we will sometimes refer to  $Z_D$  as the Rosenblatt distribution itself.

By Proposition 4.6.2, the Rosenblatt distribution has cumulants  $\chi_1 = 0$  and

$$\chi_p := \chi_p(Z_D) = 2^{-1}(p-1)!C_{p,2,H_0}S_p, \quad p \geq 2, \quad (4.7.2)$$

and the characteristic function

$$\mathbb{E}e^{i\theta Z_D} = \exp \left\{ \frac{1}{2} \sum_{p=2}^{\infty} \frac{i^p \theta^p C_{p,2,H_0} S_p}{p} \right\} = \exp \left\{ \sum_{p=2}^{\infty} \frac{i^p \theta^p}{p!} \chi_p \right\}, \quad (4.7.3)$$

where

$$S_p = \int_0^1 ds_1 \dots \int_0^1 ds_p |s_1 - s_2|^{-D} |s_2 - s_3|^{-D} \dots |s_p - s_1|^{-D}. \quad (4.7.4)$$

Each moment  $\mu_p = \mathbb{E}Z_D^p$ ,  $p \geq 2$ , can be expressed as a polynomial using the cumulants  $\chi_k$ ,  $k = 1, \dots, p$ , and the *complete Bell polynomials*

$$\mu_p = B_p(0, \chi_2, \dots, \chi_p), \quad (4.7.5)$$

as noted, for example, in Peccati and Taqqu [797], Proposition 3.3.1. They can also be computed recursively (e.g., Smith [914]).

Thus, in order to compute any moment or cumulant, it is necessary to compute the multiple integrals  $S_k$ . The first two can be computed directly as

$$S_2 = \frac{1}{(1-D)(1-2D)}, \quad S_3 = \frac{2}{(1-D)(2-3D)} B(1-D, 1-D) \quad (4.7.6)$$

(Exercise 4.6). For  $p \geq 4$ , a closed form expression for  $S_p$  is not known, which means they must be computed numerically. Computing the multiple integrals directly is intractable due to the increasing number of singularities in the integrand. This difficulty can be circumvented through the following procedure.

Denote  $L^2(0, 1) = L^2((0, 1), dx)$  and define the integral operator  $\mathcal{K}_D : L^2(0, 1) \rightarrow L^2(0, 1)$  as

$$(\mathcal{K}_D f)(x) = \int_0^1 |x - u|^{-D} f(u) du \quad (4.7.7)$$

and then the sequence of functions  $G_{k,D} \in L^2(0, 1)$ ,  $k \geq 1$ , recursively as follows:

$$G_{1,D}(x) = \frac{(1-x)^{-D}}{\sqrt{1-D}}, \quad G_{k,D}(x) = (\mathcal{K}_D G_{k-1,D})(x), \quad k \geq 2. \quad (4.7.8)$$

Then, we have the following alternative way to express  $S_p$ . See Proposition 2.1 in Veillette and Taqqu [979] for a proof. For  $f, g \in L^2(0, 1)$ ,  $(f, g)_{L^2(0,1)} = \int_0^1 f(x)g(x)dx$  denotes their inner product.

**Proposition 4.7.2** *Let  $\eta$  and  $\nu$  be any two positive integers such that  $\eta + \nu = p$ . Then,*

$$S_p = (G_{\eta,D}, G_{\nu,D})_{L^2(0,1)}. \quad (4.7.9)$$

To minimize the number of integrals one needs to compute, it makes sense to choose  $\eta = \nu = \frac{p}{2}$  if  $p$  is even, and  $\eta = \frac{p+1}{2}$  and  $\nu = \frac{p-1}{2}$  if  $p$  is odd. Proposition 4.7.2 thus reduces the problem of computing a  $p$ -dimensional integral into computing  $\lceil \frac{p}{2} \rceil + 1$  one-dimensional integrals. One can also show that

$$G_{2,D}(x) = \frac{x^{1-D}}{(1-D)^{3/2}} {}_2F_1(D, 1, 2-D, x) + \frac{(1-x)^{1-2D} B(1-D, 1-D)}{\sqrt{1-D}},$$

where  ${}_2F_1(a, b, c, x)$  is the so-called *Gauss hypergeometric function* (see (6.2.16) and (6.2.17) in Chapter 6). The function  ${}_2F_1(a, b, c; x)$  is bounded for  $x \in (0, 1)$  as long as  $c > a + b$  (Lebedev [601], Section 9), which is true in this case. This implies that unlike  $G_{1,D}$ ,  $G_{2,D}$  is a bounded function on  $(0, 1)$ , since  $0 < D < 1/2$ .

In Section C of Veillette and Taquq [980], the authors outline a technique for computing  $S_p$  numerically based on Proposition 4.7.2, and tabulate the first eight cumulants and moments of the Rosenblatt distribution for various values of  $D$ .

By Lemma 4.5.2, the Rosenblatt distribution can be represented as

$$Z_D = \sum_{k=1}^{\infty} \lambda_k(D) (\xi_k^2 - 1), \quad (4.7.10)$$

where  $\xi_k, k \geq 1$ , are i.i.d.  $\mathcal{N}(0, 1)$  random variables, and  $\lambda_k, k \geq 1$ , are the eigenvalues (in the decreasing order) of the operator  $A_D : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$  defined by

$$(A_D f)(v) = a_{2,H_0} \int_{\mathbb{R}} \left\{ \int_0^1 (s-u)_+^{-\frac{1+D}{2}} (s-v)_+^{-\frac{1+D}{2}} ds \right\} f(u) du, \quad (4.7.11)$$

where we used the representation (4.2.4) with  $t = 1$  of the Rosenblatt distribution. The next result provides a convenient alternative characterization of the eigenvalues  $\lambda_k(D), k \geq 1$ .

**Proposition 4.7.3** *The eigenvalues  $\lambda_k(D), k \geq 1$ , are the same as those of the operator  $(\sigma_D)\mathcal{K}_D$  in (4.7.7), where*

$$\sigma_D = \left( \frac{1}{2} (1-D)(1-2D) \right)^{1/2}. \quad (4.7.12)$$

*Proof* We shall prove only one direction of the statement, namely, that the eigenvalues  $\lambda_k(D), k \geq 1$ , are also those of the operator  $(\sigma_D)\mathcal{K}_D$ . See Proposition 3.1 in Veillette and Taquq [979] for a slightly different and complete proof. We should also note that the arguments below could be understood easier using fractional calculus of Chapters 6 and 7. But to keep the presentation self-contained, we will not use here any notions or results of fractional calculus.

If  $\lambda_k(D)$  is an eigenvalue of the operator  $A_D$ , there is a function  $f \in L^2(\mathbb{R})$  such that

$$a_{2,H_0} \int_{\mathbb{R}} \left\{ \int_0^1 (s-u)_+^{-\frac{1+D}{2}} (s-v)_+^{-\frac{1+D}{2}} ds \right\} f(u) du = \lambda_k(D) f(v)$$

or

$$a_{2,H_0} \int_0^1 (s-v)_+^{-\frac{1+D}{2}} \left\{ \int_{\mathbb{R}} (s-u)_+^{-\frac{1+D}{2}} f(u) du \right\} ds = \lambda_k(D) f(v).$$

After multiplying both sides of the last relation by  $(z-v)_+^{-\frac{1+D}{2}}$ ,  $z \in (0, 1)$ , and integrating over  $v \in \mathbb{R}$ , we obtain that

$$a_{2,H_0} \int_0^1 \left\{ \int_{\mathbb{R}} (z-v)_+^{-\frac{1+D}{2}} (s-v)_+^{-\frac{1+D}{2}} dv \right\} g(s) ds = \lambda_k(D) g(z),$$

where  $g(z) = \int_{\mathbb{R}} (z-v)_+^{-\frac{1+D}{2}} f(v) dv$ . The inner integral over  $v$  above can be evaluated explicitly. Indeed, for example, for  $z < s$ , by making the changes of variables  $z-v = (s-z)y$ , we have

$$\begin{aligned} & \int_{\mathbb{R}} (z-v)_+^{-\frac{1+D}{2}} (s-v)_+^{-\frac{1+D}{2}} dv (s-z)^{-D} \int_0^\infty y^{-\frac{1+D}{2}} (1+y)^{-\frac{1+D}{2}} dy \\ &= (s-z)^{-D} B\left(\frac{1-D}{2}, D\right), \end{aligned}$$

where we also used (2.2.12). Observe from (4.2.8) that

$$a_{2,H_0} = \left(\frac{1}{2}(1-D)(1-2D)\right)^{1/2} \frac{1}{B((1-D)/2, D)} = \frac{\sigma_D}{B((1-D)/2, D)}.$$

It follows that

$$\sigma_D \int_0^1 |s-z|^{-D} g(s) ds = \lambda_k(D) g(z),$$

that is,  $\lambda_k(D)$  is an eigenvalue of the operator  $(\sigma_D)\mathcal{K}_D$  as well.  $\square$

**Remark 4.7.4** The following asymptotic relationships for the eigenvalues  $\lambda_k(D)$  can be established (Theorem 3.2 in Veillette and Taquq [979]). For  $0 < r < 1$ ,

$$\lambda_k(D) = C(D) k^{D-1} \left(1 + o\left(\frac{1}{k^r}\right)\right), \quad (4.7.13)$$

where

$$C(D) = \frac{2}{\pi^{1-D}} \sigma_D \Gamma(1-D) \sin\left(\frac{\pi D}{2}\right). \quad (4.7.14)$$

Moreover, the series

$$\sum_{k=1}^{\infty} (\lambda_k(D) - C(D) k^{D-1})$$

converges and equals

$$\sum_{k=1}^{\infty} (\lambda_k(D) - C(D) k^{D-1}) = -2^{1-D} \sigma_D \zeta(D), \quad (4.7.15)$$

where  $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$  denotes the Riemann zeta function.

Further insight on the Rosenblatt distribution can be gained by regarding it as an infinitely divisible distribution. Recall that the characteristic function of an infinitely divisible distribution  $X$  with  $\mathbb{E}X^2 < \infty$  can be written in the following form:

$$\phi(\theta) = \mathbb{E}e^{i\theta X} = \exp \left\{ ia\theta - \frac{1}{2}b^2\theta^2 + \int_{-\infty}^{\infty} (e^{i\theta u} - 1 - i\theta u)v(du) \right\},$$

where  $a \in \mathbb{R}$ ,  $b > 0$  and  $v$  is a positive measure on  $\mathbb{R} \setminus \{0\}$  with the property that  $\int \min\{u^2, 1\}v(du) < \infty$ . This is known as the Lévy–Khintchine representation of  $X$ . Since the chi-square distribution is infinitely divisible, it is not surprising in light of (4.7.10) that the Rosenblatt distribution is also infinitely divisible.

The next result provides the Lévy–Khintchine representation of the Rosenblatt distribution.

**Theorem 4.7.5** *Let  $Z_D$  have the Rosenblatt distribution with  $0 < D < 1/2$  and let*

$$\lambda(D)^{-1} = (\lambda_1(D)^{-1}, \lambda_2(D)^{-1}, \dots)$$

*be the sequence of the inverses of the eigenvalues associated to the integral operator  $\sigma_D \mathcal{K}_D$  defined in (4.7.7) and (4.7.12). Then, the characteristic function of  $Z_D$  can be written as*

$$\phi(\theta) = \mathbb{E}e^{i\theta Z_D} = \exp \left\{ \int_0^\infty (e^{i\theta u} - 1 - i\theta u)v_D(u)du \right\}, \quad (4.7.16)$$

where  $v_D$  is supported on  $(0, \infty)$  and is given by

$$v_D(u) = \frac{1}{2u} \sum_{n=1}^{\infty} e^{-u/(2\lambda_n(D))}, \quad u > 0. \quad (4.7.17)$$

Moreover,  $v_D$  has the following asymptotic forms as  $u \rightarrow 0^+$  and  $u \rightarrow \infty$ ,

$$v_D(u) \sim \frac{2^{\frac{D}{1-D}} C(D)^{\frac{1}{1-D}}}{(1-D)} \Gamma\left(\frac{1}{1-D}\right) u^{\frac{D-2}{1-D}}, \quad u \rightarrow 0^+, \quad (4.7.18)$$

$$v_D(u) \sim \frac{e^{-u/(2\lambda_1)}}{2u}, \quad u \rightarrow \infty, \quad (4.7.19)$$

where  $C(D)$  is defined in (4.7.14).

*Proof* Let

$$Z_D^{(M)} = \sum_{n=1}^M \lambda_n(D) (\xi_n^2 - 1).$$

We have  $Z_D^{(M)} \xrightarrow{d} Z_D$ , and since  $Z_D^{(M)}$  is a sum of shifted i.i.d. chi-squared distributions, we can use the Lévy–Khintchine representation of a chi-square (Applebaum [40], Example 1.3.22), which is a gamma distribution with shape parameter  $1/2$  and scale parameter  $2$ :

$$\mathbb{E}e^{i\theta Z_D^{(M)}} = \prod_{n=1}^M e^{i\theta \lambda_n(D) (\xi_n^2 - 1)} = \prod_{n=1}^M \exp \left\{ -i\theta \lambda_n(D) + \int_0^\infty (e^{i\theta u} - 1) \frac{e^{-u/(2\lambda_n(D))}}{2u} du \right\}. \quad (4.7.20)$$



Using  $(1/2) \int_0^\infty e^{-u/(2\lambda)} du = \lambda$ , (4.7.20) can be rewritten as

$$\begin{aligned} & \prod_{n=1}^M \exp \left\{ \int_0^\infty (e^{i\theta u} - 1 - i\theta u) \frac{e^{-u/(2\lambda_n(D))}}{2u} du \right\} \\ &= \exp \left\{ \int_0^\infty (e^{i\theta u} - 1 - i\theta u) \frac{1}{2u} G_{\lambda(D)^{-1}}^{(M)}(e^{-u/2}) du \right\}, \end{aligned}$$

where

$$G_{\lambda(D)^{-1}}^{(M)}(x) = \sum_{n=1}^M x^{\lambda_n(D)^{-1}}.$$

Now, we let  $M \rightarrow \infty$ . In order to justify passing the limit through the integral, notice that

$$\left| (e^{i\theta u} - 1 - i\theta u) \frac{1}{2u} G_{\lambda(D)^{-1}}^{(M)}(e^{-u/2}) \right| \leq \frac{\theta^2}{4} u G_{\lambda(D)^{-1}}^{(M)}(e^{-u/2}) \leq \frac{\theta^2}{4} u G_{\lambda(D)^{-1}}(e^{-u/2}), \quad (4.7.21)$$

where we have used the identity  $|e^{iz} - 1 - z| \leq z^2/2$  for  $z \in \mathbb{R}$  and the definition (4.7.24) for the function  $G_{\lambda(D)^{-1}}(x)$ . Notice that the last bound in (4.7.21) is continuous for  $0 < u < \infty$ , and by (4.7.13) together with Lemma 4.7.6 below using  $\alpha = 1 - D$  and  $\beta = C(D)^{-1}$ , we have

$$u G_{\lambda(D)^{-1}}(e^{-u/2}) \sim u e^{-u/(2\lambda_1)}, \quad \text{as } u \rightarrow \infty, \quad (4.7.22)$$

and

$$u G_{\lambda(D)^{-1}}(e^{-u/2}) \sim C' u (1 - e^{-u/2})^{-1/(1-D)} \sim C'' u^{\frac{D}{1-D}}, \quad \text{as } u \rightarrow 0^+, \quad (4.7.23)$$

for some constants  $C'$  and  $C''$ . Since  $0 < \frac{D}{1-D} < 1$ , (4.7.22) and (4.7.23) imply that the last bound in (4.7.21) is integrable on  $(0, \infty)$ , and hence the dominated convergence theorem applies and

$$\mathbb{E} e^{i\theta Z_D^{(M)}} \rightarrow \mathbb{E} e^{i\theta Z_D} = \exp \left\{ \int_0^\infty (e^{i\theta u} - 1 - i\theta u) \frac{1}{2u} G_{\lambda(D)^{-1}}(e^{-u/2}) du \right\},$$

which verifies (4.7.16).

The final assertions (4.7.18) and (4.7.19) also follow from (4.7.13) and Lemma 4.7.6 below with  $\alpha = 1 - D$  and  $\beta = C(D)^{-1}$ , since these imply

$$\begin{aligned} \frac{1}{2u} G_{\lambda(D)^{-1}}(e^{-u/2}) &\sim \frac{1}{2u} \frac{C(D)^{-\frac{1}{1-D}}}{(1-D)} \Gamma\left(\frac{1}{1-D}\right) \left(\frac{u}{2}\right)^{-\frac{1}{1-D}} \\ &= \frac{2^{\frac{D}{1-D}} C(D)^{\frac{1}{1-D}}}{(1-D)} \Gamma\left(\frac{1}{1-D}\right) u^{\frac{D-2}{1-D}}, \quad u \rightarrow 0^+, \end{aligned}$$

and

$$\frac{1}{2u} G_{\lambda(D)^{-1}}(e^{-u/2}) \sim \frac{1}{2u} e^{-u/(2\lambda_1(D))}, \quad u \rightarrow \infty.$$

This concludes the proof.  $\square$

The next auxiliary lemma was used in the proof of Theorem 4.7.5 above. See Lemma 4.1 in Veillette and Taqqu [1979] for a proof. Given any positive, increasing sequence  $\mathbf{c} = \{c_n\}_{n=1}^\infty$  such that  $\sum_{n=1}^\infty 1/c_n^2 < \infty$ , define the function  $G_{\mathbf{c}}(x)$  for  $0 < x < 1$  as

$$G_{\mathbf{c}}(x) = \sum_{n=1}^{\infty} x^{c_n}. \quad (4.7.24)$$

Since  $\mathbf{c}^{-1} = \{c_1^{-1}, c_2^{-1}, \dots\} \in \ell^2(\mathbb{Z})$ , we have  $c_n \rightarrow \infty$  and thus this series converges for all  $x \in (0, 1)$  since  $\log x < 0$  and hence for  $n$  large enough, one has  $x^{c_n} = e^{c_n \log(x)} \leq c_n^{-2}$ . Notice that  $G_{\mathbf{c}}(0) = 0$ ,  $G_{\mathbf{c}}(x) \rightarrow \infty$  as  $x \rightarrow 1$  and  $G_{\mathbf{c}}(x)$  is a continuous function for all  $x \in (0, 1)$ .

**Lemma 4.7.6** *Suppose  $\mathbf{c}$  is a positive strictly increasing sequence such that  $c_n \sim \beta n^\alpha$ , as  $n \rightarrow \infty$ , for some  $1/2 < \alpha < 1$  and constant  $\beta > 0$ . Then,*

$$G_{\mathbf{c}}(x) \sim x^{c_1}, \quad \text{as } x \rightarrow 0, \quad (4.7.25)$$

$$G_{\mathbf{c}}(x) \sim \frac{1}{\alpha \beta^{\frac{1}{\alpha}}} \Gamma\left(\frac{1}{\alpha}\right) (1-x)^{-\frac{1}{\alpha}}, \quad \text{as } x \rightarrow 1. \quad (4.7.26)$$

Understanding the Lévy measure of a distribution has some immediate implications pertaining to its probability density function and distribution function. We state three such results in the following corollary. See Veillette and Taqqu [1979] for their proofs, as well as Exercise 4.7.

**Corollary 4.7.7** *Let  $Z_D$  denote the Rosenblatt distribution. Then:*

- (i) *The probability density function of  $Z_D$  is infinitely differentiable with all derivatives tending to 0 as  $|x| \rightarrow \infty$ .*
- (ii) *Left tail: for  $x > 0$ ,*

$$\mathbb{P}(Z_D < -x) \leq e^{-\frac{x^2}{2}}.$$

- (iii) *Right tail: for  $\alpha > 0$ ,*

$$\lim_{u \rightarrow \infty} \frac{\mathbb{P}(Z_D > u + \alpha)}{\mathbb{P}(Z_D > u)} = e^{-\frac{\alpha}{2\lambda_1(D)}},$$

where  $\lambda_1(D)$  is the largest eigenvalue of  $\sigma_D \mathcal{K}_D$  defined in (4.7.7) and (4.7.12).

The representation (4.7.10) can also be used to compute numerically the cumulative distribution function (CDF) and the probability density function (PDF) of the Rosenblatt distribution. For  $M \geq 1$ , define  $X_M$  and  $Y_M$  as

$$Z_D = \sum_{n=1}^{M-1} \lambda_n (\xi_n^2 - 1) + \sum_{n=M}^{\infty} \lambda_n (\xi_n^2 - 1) := X_M + Y_M. \quad (4.7.27)$$

Notice that  $Y_M$  has mean 0 and variance

$$\sigma_M^2 := \mathbb{E}Y_M^2 = 2 \sum_{n=M}^{\infty} \lambda_n^2. \quad (4.7.28)$$

Notice from (4.7.13) that, as  $M \rightarrow \infty$ ,

$$\sigma_M^2 \sim 2C(D)^2 \sum_{n=M}^{\infty} n^{2D-2} \sim 2C(D)^2 (1-2D)^{-1} M^{2D-1}, \quad (4.7.29)$$

where we approximated the sum  $\sum_{n=M}^{\infty} \lambda_n^2$  with an integral. This suggests that  $X_M$  alone is not a very good approximation of  $Z_D$  since this variance tends to 0 slowly with  $M$ , especially when  $D$  is close to  $1/2$ . As an alternative, as seen below,  $Y_M$  can be approximated by a normal distribution as  $M \rightarrow \infty$ . By taking advantage of this property, we can obtain accurate approximations of the distribution of  $Z_D$ .

A normal approximation of  $Y_M$  based on an Edgeworth expansion was established in Veillette and Taquq [978]. Let  $\kappa_{k,M}$  be the normalized cumulants of  $Y_M$ . These are given by

$$\kappa_{k,M} = 2^{k-1} (k-1)! \sigma_M^{-k} \sum_{n=M}^{\infty} \lambda_n(D)^k. \quad (4.7.30)$$

Notice that from the asymptotics (4.7.13) of  $\lambda_n(D)$ ,

$$\begin{aligned} \kappa_{k,M} &\sim 2^{k-1} (k-1)! \cdot 2^{-k/2} (1-2D)^{k/2} M^{k/2-kD} \cdot \int_M^{\infty} n^{k(D-1)} dn \\ &= \frac{(k-1)!}{2} \frac{(2-4D)^{k/2}}{k-kD-1} M^{1-k/2}. \end{aligned}$$

Let also  $\Phi$  and  $\phi$  be the CDF and PDF of a standard normal distribution, respectively. The CDF of the tail  $Y_M$  satisfies

$$\mathbb{P}(\sigma_M^{-1} Y_M \leq x) = \Phi(x) - \phi(x) \left\{ \sum_{\eta(N)} \left[ \prod_{m=1}^N \frac{1}{k_m!} \left( \frac{\kappa_{m,M}}{m!} \right)^{k_m} \right] H_{\zeta(k_3, \dots, k_N)}(x) \right\} + O\left(M^{-\frac{N-1}{2}}\right), \quad (4.7.31)$$

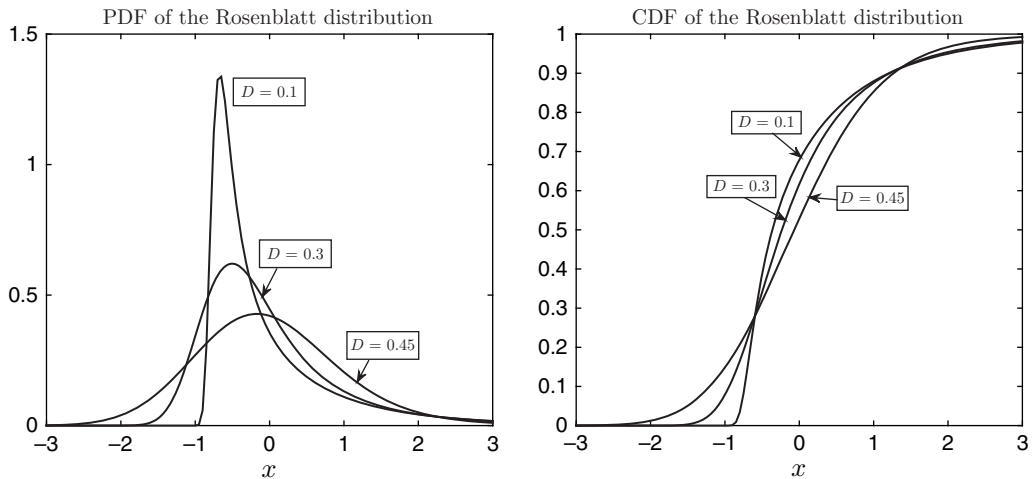
where  $H_n$  denotes the Hermite polynomial of order  $n$ ,  $\kappa_{k,M}$  is defined in (4.7.30),  $\eta(N)$  denotes all  $k_3, k_4, \dots, k_N$  such that

$$1 \leq k_3 + 2k_4 + \dots + (N-2)k_N \leq N-2$$

and

$$\zeta(k_3, \dots, k_N) = 3k_3 + 4k_4 + \dots + Nk_N - 1.$$

The relation (4.7.31) can be used to evaluate the CDF of the tail  $Y_M$ . For  $X_M$ , methods exist already for accurately computing the CDF of a finite sum of chi-squared distributions, see for instance methods based on Laplace transform inversion (Veillette and Taquq [977], Castaño-Martínez and López-Blázquez [207]), or Fourier transform inversion (Abate and Whitt [4]). Both tasks require computing the eigenvalues  $\lambda_k(D)$ . The computation of these eigenvalues, as those of the operator  $(\sigma_D)\mathcal{K}_D$ , is discussed in Veillette and Taquq [979, 980].



**Figure 4.13** Plots of the PDF and CDF of  $Z_D$  for various  $D$ . The CDF with the steepest slope and the PDF with the highest mode correspond to  $D = 0.1$ .

Finally, in Figure 4.13, we present the plots of the CDF and PDF of the Rosenblatt distribution for several values of  $D$  (with  $M = 50$  and  $N = 5$ ). See Veillette and Taqqu [979, 980] for further technical details on their computation. A curious observation from Figure 4.13 is that the CDFs seem to intersect at the same point for the different considered values of  $D$  (Exercise 4.8).

#### 4.8 CDF of the Rosenblatt Distribution

In Figure 4.13, we presented the plots of the PDF and CDF of the Rosenblatt distribution, that is, of the variable  $Z_D = Z_H^{(2)}(1)$ , where  $D = 1 - H \in (0, 1/2)$  and  $\{Z_H^{(2)}(t)\}_{t \in \mathbb{R}}$ ,  $H \in (1/2, 1)$ , is the standardized Rosenblatt process (see Definition 4.7.1). This process can appear as a limit in statistical problems. In order to obtain confidence intervals, it is necessary to compute numerically CDF of the random variable  $Z_D$ . Note that since  $\{Z_H^{(2)}(t)\}_{t \in \mathbb{R}}$  is  $H$ -self-similar, one obtains the CDF of  $Z_H(t)$ ,  $t > 0$ , by using the relation  $\mathbb{P}(Z_H(t) < z) = \mathbb{P}(Z_H(1) < t^{-H}z)$ .

Table 4.1 gives the CDF and Table 4.2 the quantiles of the random variable  $Z_D$ . They were computed numerically using the approximation described in Veillette and Taqqu [979].

#### 4.9 Generalized Hermite and Related Processes

We describe first an interesting class of processes generalizing the Hermite processes. Throughout this section, we use the following notation: for  $u = (u_1, \dots, u_k) \in \mathbb{R}^k$  and  $s \in \mathbb{R}$ , we write

$$s + u = (s + u_1, \dots, s + u_k), \quad su = (su_1, \dots, su_k), \quad \{s > u\} = \{s > u_1, \dots, s > u_k\}, \quad (4.9.1)$$

Table 4.1 Values of the CDF  $F(x) = \mathbb{P}(Z_D \leq x)$  for various values of  $x$  and  $D \in (0, 1/2)$ , for the standardized Rosenblatt distribution. An entry of 0.0000 means the CDF takes a value less than  $10^{-4}$ .

| $x$  | $D = 0.1$ | $D = 0.2$ | $D = 0.3$ | $D = 0.4$ | $D = 0.45$ | $x$ | $D = 0.1$ | $D = 0.2$ | $D = 0.3$ | $D = 0.4$ | $D = 0.45$ |
|------|-----------|-----------|-----------|-----------|------------|-----|-----------|-----------|-----------|-----------|------------|
| -2.5 | 0.0000    | 0.0000    | 0.0000    | 0.0002    | 0.0020     | 1.0 | 0.8797    | 0.8790    | 0.8755    | 0.8645    | 0.8532     |
| -2.4 | 0.0000    | 0.0000    | 0.0000    | 0.0004    | 0.0030     | 1.1 | 0.8901    | 0.8899    | 0.8879    | 0.8805    | 0.8725     |
| -2.3 | 0.0000    | 0.0000    | 0.0000    | 0.0007    | 0.0043     | 1.2 | 0.8996    | 0.8997    | 0.8989    | 0.8946    | 0.8897     |
| -2.2 | 0.0000    | 0.0000    | 0.0000    | 0.0013    | 0.0062     | 1.3 | 0.9081    | 0.9086    | 0.9087    | 0.9071    | 0.9047     |
| -2.1 | 0.0000    | 0.0000    | 0.0000    | 0.0022    | 0.0088     | 1.4 | 0.9159    | 0.9166    | 0.9174    | 0.9180    | 0.9179     |
| -2   | 0.0000    | 0.0000    | 0.0000    | 0.0037    | 0.0122     | 1.5 | 0.9230    | 0.9238    | 0.9253    | 0.9276    | 0.9294     |
| -1.9 | 0.0000    | 0.0000    | 0.0000    | 0.0059    | 0.0167     | 1.6 | 0.9294    | 0.9304    | 0.9323    | 0.9360    | 0.9393     |
| -1.8 | 0.0000    | 0.0000    | 0.0001    | 0.0093    | 0.0225     | 1.7 | 0.9353    | 0.9364    | 0.9386    | 0.9434    | 0.9479     |
| -1.7 | 0.0000    | 0.0000    | 0.0004    | 0.0142    | 0.0299     | 1.8 | 0.9406    | 0.9418    | 0.9443    | 0.9499    | 0.9554     |
| -1.6 | 0.0000    | 0.0000    | 0.0011    | 0.0210    | 0.0392     | 1.9 | 0.9455    | 0.9467    | 0.9494    | 0.9556    | 0.9617     |
| -1.5 | 0.0000    | 0.0000    | 0.0030    | 0.0303    | 0.0506     | 2.0 | 0.9499    | 0.9512    | 0.9541    | 0.9606    | 0.9672     |
| -1.4 | 0.0000    | 0.0000    | 0.0070    | 0.0425    | 0.0645     | 2.1 | 0.9540    | 0.9553    | 0.9582    | 0.9650    | 0.9719     |
| -1.3 | 0.0000    | 0.0000    | 0.0148    | 0.0582    | 0.0810     | 2.2 | 0.9577    | 0.9590    | 0.9620    | 0.9689    | 0.9758     |
| -1.2 | 0.0000    | 0.0008    | 0.0282    | 0.0777    | 0.1003     | 2.3 | 0.9611    | 0.9624    | 0.9654    | 0.9724    | 0.9793     |
| -1.1 | 0.0000    | 0.0052    | 0.0489    | 0.1014    | 0.1227     | 2.4 | 0.9643    | 0.9655    | 0.9685    | 0.9754    | 0.9822     |
| -1   | 0.0000    | 0.0205    | 0.0785    | 0.1293    | 0.1481     | 2.5 | 0.9671    | 0.9684    | 0.9713    | 0.9781    | 0.9847     |
| -0.9 | 0.0011    | 0.0554    | 0.1174    | 0.1616    | 0.1766     | 2.6 | 0.9697    | 0.9710    | 0.9739    | 0.9805    | 0.9868     |
| -0.8 | 0.0344    | 0.1134    | 0.1649    | 0.1980    | 0.2081     | 2.7 | 0.9721    | 0.9733    | 0.9762    | 0.9826    | 0.9886     |
| -0.7 | 0.1435    | 0.1893    | 0.2195    | 0.2380    | 0.2424     | 2.8 | 0.9744    | 0.9755    | 0.9783    | 0.9845    | 0.9902     |
| -0.6 | 0.2754    | 0.2737    | 0.2789    | 0.2810    | 0.2792     | 2.9 | 0.9764    | 0.9775    | 0.9802    | 0.9861    | 0.9915     |
| -0.5 | 0.3864    | 0.3575    | 0.3406    | 0.3264    | 0.3182     | 3.0 | 0.9782    | 0.9793    | 0.9819    | 0.9876    | 0.9926     |
| -0.4 | 0.4726    | 0.4349    | 0.4022    | 0.3734    | 0.3588     | 3.1 | 0.9799    | 0.9810    | 0.9835    | 0.9889    | 0.9936     |
| -0.3 | 0.5402    | 0.5035    | 0.4619    | 0.4211    | 0.4007     | 3.2 | 0.9815    | 0.9825    | 0.9849    | 0.9901    | 0.9944     |
| -0.2 | 0.5947    | 0.5630    | 0.5180    | 0.4687    | 0.4433     | 3.3 | 0.9830    | 0.9839    | 0.9862    | 0.9911    | 0.9952     |
| -0.1 | 0.6400    | 0.6140    | 0.5698    | 0.5154    | 0.4861     | 3.4 | 0.9843    | 0.9852    | 0.9874    | 0.9921    | 0.9958     |
| 0    | 0.6784    | 0.6577    | 0.6169    | 0.5605    | 0.5285     | 3.5 | 0.9855    | 0.9864    | 0.9885    | 0.9929    | 0.9963     |
| 0.1  | 0.7115    | 0.6952    | 0.6591    | 0.6036    | 0.5700     | 3.6 | 0.9866    | 0.9875    | 0.9895    | 0.9936    | 0.9968     |
| 0.2  | 0.7403    | 0.7277    | 0.6967    | 0.6441    | 0.6102     | 3.7 | 0.9877    | 0.9885    | 0.9904    | 0.9943    | 0.9972     |
| 0.3  | 0.7655    | 0.7558    | 0.7300    | 0.6818    | 0.6488     | 3.8 | 0.9886    | 0.9894    | 0.9912    | 0.9949    | 0.9975     |
| 0.4  | 0.7878    | 0.7805    | 0.7594    | 0.7165    | 0.6853     | 3.9 | 0.9895    | 0.9902    | 0.9919    | 0.9954    | 0.9978     |
| 0.5  | 0.8077    | 0.8021    | 0.7853    | 0.7482    | 0.7196     | 4.0 | 0.9903    | 0.9910    | 0.9926    | 0.9959    | 0.9981     |
| 0.6  | 0.8253    | 0.8213    | 0.8081    | 0.7769    | 0.7514     | 4.1 | 0.9910    | 0.9917    | 0.9933    | 0.9963    | 0.9983     |
| 0.7  | 0.8412    | 0.8382    | 0.8282    | 0.8027    | 0.7807     | 4.2 | 0.9917    | 0.9924    | 0.9938    | 0.9967    | 0.9985     |
| 0.8  | 0.8554    | 0.8534    | 0.8459    | 0.8258    | 0.8074     | 4.3 | 0.9924    | 0.9930    | 0.9943    | 0.9970    | 0.9986     |
| 0.9  | 0.8682    | 0.8669    | 0.8616    | 0.8463    | 0.8315     | 4.4 | 0.9929    | 0.9935    | 0.9948    | 0.9973    | 0.9988     |
|      |           |           |           |           |            | 4.5 | 0.9935    | 0.9940    | 0.9953    | 0.9976    | 0.9989     |

with analogous definitions when  $+$  is replaced by  $-$ ,  $>$  is replaced by  $\geq$ ,  $<$ , etc. Throughout this section, we also let  $I_k$  denote a multiple integral of order  $k$  with respect to a Gaussian random measure on  $\mathbb{R}$  having the Lebesgue control measure.

We shall use the following elementary result which provides a general way to construct self-similar processes with stationary increments living in the  $k$ th Wiener chaos; that is, expressed through the multiple integral  $I_k$ . The proof of the result is left as Exercise 4.9.

Table 4.2 Various quantiles of the Rosenblatt distribution for selected values of  $D$ .

| Quantile | $D = 0.1$ | $D = 0.2$ | $D = 0.3$ | $D = 0.4$ | $D = 0.45$ |
|----------|-----------|-----------|-----------|-----------|------------|
| 0.01     | -0.8472   | -1.0567   | -1.3546   | -1.7838   | -2.0603    |
| 0.025    | -0.8142   | -0.9827   | -1.7669   | -1.5536   | -1.7639    |
| 0.05     | -0.7808   | -0.9122   | -1.0958   | -1.3493   | -1.5051    |
| 0.10     | -0.7340   | -0.8201   | -0.9419   | -1.1053   | -1.6462    |
| 0.25     | -0.6200   | -0.6277   | -0.6479   | -0.6713   | -0.6789    |
| 0.50     | -0.3622   | -0.3055   | -0.2329   | -0.1332   | -0.0673    |
| 0.75     | 0.2370    | 0.2781    | 0.3666    | 0.5059    | 0.5955     |
| 0.90     | 1.2047    | 1.2031    | 1.2110    | 1.2417    | 1.2673     |
| 0.95     | 2.0015    | 1.9726    | 1.9114    | 1.8022    | 1.7262     |
| 0.975    | 2.8312    | 2.7759    | 2.6483    | 2.3858    | 2.1774     |
| 0.99     | 3.9618    | 3.8718    | 3.6579    | 3.1909    | 2.7892     |

**Proposition 4.9.1** Let  $H \in (0, 1)$ . Suppose that  $\{h_t(\cdot), t \in \mathbb{R}\}$  is a family of functions defined on  $\mathbb{R}^k$  satisfying:

1.  $h_t \in L^2(\mathbb{R}^k)$ ;
2. For all  $\lambda > 0$ , there is  $\beta \neq 0$  such that  $h_{\lambda t}(u) = \lambda^{H+k\beta/2} h_t(\lambda^\beta u)$  for a.e.  $u \in \mathbb{R}^k$  and all  $t \in \mathbb{R}$ ;
3. For all  $s > 0$ , there exists  $a \in \mathbb{R}^k$  such that  $h_{t+s}(u) - h_t(u) = h_s(u + ta)$  for a.e.  $u \in \mathbb{R}^k$  and all  $t > 0$ .

Then,

$$Z(t) := I_k(h_t) = \int_{\mathbb{R}^k}' h_t(u_1, \dots, u_k) B(du_1) \dots B(du_k),$$

where  $B(du)$  is a Gaussian random measure on  $\mathbb{R}$  with the Lebesgue control measure  $du$ , is an  $H$ -self-similar process with stationary increments.

The following kernel functions will be used in the sequel. We let  $\mathbb{R}_+ = (0, \infty)$ .

**Definition 4.9.2** A nonzero measurable function  $g : \mathbb{R}_+^k \rightarrow \mathbb{R}$  is called a *generalized Hermite kernel* if it satisfies:

$$g(\lambda u) = \lambda^\alpha g(u), \quad \lambda > 0, u \in \mathbb{R}_+^k, \quad (4.9.2)$$

where

$$\alpha \in \left( -\frac{k+1}{2}, -\frac{k}{2} \right) \quad (4.9.3)$$

and

$$\int_{\mathbb{R}_+^k} |g(u)g(1+u)| du < \infty. \quad (4.9.4)$$

Note that (4.9.3) is equivalent to

$$0 < 2\alpha + k + 1 < 1. \quad (4.9.5)$$

The next result will allow using generalized Hermite kernels in integrands for multiple integrals.

**Proposition 4.9.3** *Let  $g$  be a generalized Hermite kernel on  $\mathbb{R}_+^k$  defined in Definition 4.9.2. Then,*

$$h_t(u) = \int_0^t g(s-u) 1_{\{s>u\}} ds \quad (4.9.6)$$

*is well-defined and belongs to  $L^2(\mathbb{R}^k)$ , for all  $t \in \mathbb{R}$ . (Note also that the convention (4.9.1) is used in (4.9.6).)*

*Proof* To check that  $h_t \in L^2(\mathbb{R}^k)$ , we write

$$\int_{\mathbb{R}^k} h_t(u)^2 du = \int_{\mathbb{R}^k} du \int_0^t \int_0^t ds_1 ds_2 g(s_1-u) g(s_2-u) 1_{\{s_1>u\}} 1_{\{s_2>u\}}.$$

It is enough to check that the right-hand side is finite with the absolute value of the integrand. Observe that, by symmetry of  $s_1$  and  $s_2$ , the right-hand side with the absolute value of the integrand is

$$\begin{aligned} & 2 \int_0^t ds_1 \int_{s_1}^t ds_2 \int_{\mathbb{R}^k} du |g(s_1-u) g(s_2-u)| 1_{\{s_1-u>0\}} \\ &= 2 \int_0^t ds \int_0^{t-s} dv \int_{\mathbb{R}_+^k} dw |g(w) g(v+w)| \\ &= 2 \int_0^t ds \int_0^{t-s} dv \int_{\mathbb{R}_+^k} dy v^k |g(vy) g(v+vy)| \\ &= 2 \int_0^t ds \int_0^{t-s} dv v^{2\alpha+k} \int_{\mathbb{R}_+^k} dy |g(y) g(1+y)|, \end{aligned}$$

where we used the changes of variables  $s = s_1$ ,  $v = s_2 - s_1$ ,  $w = s_1 - u$ , and then  $w = vy$ , and the scaling relation (4.9.2). The last integral is finite by using (4.9.4) and since  $2\alpha + k + 1 > 0$  by (4.9.5).  $\square$

The following result is an elementary corollary of Propositions 4.9.1 and 4.9.3.

**Corollary 4.9.4** *Let  $g$  be a generalized Hermite kernel defined on  $\mathbb{R}_+^k$  defined in Definition 4.9.2, and  $h_t$ ,  $t \in \mathbb{R}$ , be defined in (4.9.6). Then, the process*

$$Z(t) = I_k(h_t), \quad t \in \mathbb{R}, \quad (4.9.7)$$

*is a well-defined  $H$ -self-similar process with stationary increments, where*

$$H = \alpha + \frac{k}{2} + 1 \in \left(\frac{1}{2}, 1\right). \quad (4.9.8)$$

*Proof* We check that the functions  $h_t$  satisfy Conditions 1–3 in Proposition 4.9.1. Condition 1 is satisfied by Proposition 4.9.3. To check self-similarity (Condition 2 with  $\beta = -1$ ),

$$h_{\lambda t}(u) = \int_0^{\lambda t} g(s-u)1_{\{s>u\}}ds = \lambda^{\alpha+1} \int_0^t g(r-\lambda^{-1}u)1_{\{r>\lambda^{-1}u\}}\lambda dr = \lambda^{\alpha+1}h_t(\lambda^{-1}u),$$

where the second equality uses (4.9.2). The self-similarity parameter  $H$  of  $I_k(h_t)$  is obtained from  $\alpha + 1 = H - k/2$  (Condition 2 of Proposition 4.9.1). To check stationary increments (Condition 3 of Proposition 4.9.1), for any  $t, r > 0$ ,

$$h_{t+r}(u) - h_t(u) = \int_t^{t+r} g(s-u)1_{\{s>u\}}ds = \int_0^r g(v+t-u)1_{\{v+t>u\}}dv = h_r(u-t). \quad \square$$

**Definition 4.9.5** The process  $\{Z(t)\}_{t \in \mathbb{R}}$  in (4.9.7), that is,

$$Z(t) = \int_{\mathbb{R}^k}' \left\{ \int_0^t g(s-u_1, \dots, s-u_k)1_{\{s>u_1, \dots, s>u_k\}}ds \right\} B(du_1) \dots B(du_k), \quad (4.9.9)$$

where  $B(du)$  is a Gaussian random measure on  $\mathbb{R}$  with the Lebesgue control measure  $du$  and  $g$  is a generalized Hermite kernel defined in Definition 4.9.2, is called a *generalized Hermite process*.

**Remark 4.9.6** It is known (see, e.g., Janson [529], Theorem 6.12) that if a random variable  $X$  belongs to the  $k$ th Wiener chaos, then there exist  $a, b, t_0 > 0$  such that for  $t \geq t_0$ ,

$$\exp\{-at^{2/k}\} \leq \mathbb{P}(|X| > t) \leq \exp\{-bt^{2/k}\}.$$

This shows that the generalized Hermite processes of different orders must necessarily have different laws, and the higher the order gets, the heavier the tail of the marginal distribution becomes, while they all have moments of any order. When two generalized Hermite process of the same order have the same law remains an open question.

**Remark 4.9.7** The expression (4.9.9) is a time domain representation of a generalized Hermite process, analogous to (4.2.3)–(4.2.4) for the Hermite process. There is also a spectral domain representation of a generalized Hermite process, analogous to (4.2.11) for the Hermite process. Since the function  $h_t \in L^2(\mathbb{R}^k)$ , it has the  $L^2(\mathbb{R}^k)$ –Fourier transform, which can be written as

$$\widehat{h}_t(x) = \widehat{h}_t(x_1, \dots, x_k) = \frac{e^{it(x_1+\dots+x_k)} - 1}{i(x_1 + \dots + x_k)} \widehat{g}(-x), \quad x = (x_1, \dots, x_k) \in \mathbb{R}^k, \quad (4.9.10)$$

for a function  $\widehat{g}(x)$  satisfying the scaling relation: for  $\lambda > 0$ ,  $\widehat{g}(\lambda x) = \lambda^{-\alpha-k} \widehat{g}(x)$  a.e.  $dx$  (see Propositions 3.12 and 3.14 in Bai and Taqqu [77]). In view of (B.2.16), the spectral representation of a generalized Hermite process is then given by

$$Z(t) = \frac{1}{(2\pi)^{k/2}} \int_{\mathbb{R}^k}'' \frac{e^{it(x_1+\dots+x_k)} - 1}{i(x_1 + \dots + x_k)} \widehat{g}(-x) \widehat{B}(dx_1) \dots \widehat{B}(dx_k). \quad (4.9.11)$$

We next provide several special examples of generalized Hermite processes.



**Example 4.9.8 (Hermite kernel)** The usual Hermite processes are obtained from the generalized Hermite processes by taking the generalized Hermite kernel  $g$  as (up to a constant)

$$g(u) = g(u_1, \dots, u_k) = \prod_{j=1}^k u_j^{\alpha/k}, \quad u_j > 0, \quad (4.9.12)$$

where  $\alpha/k = H_0 - 3/2$  with  $H_0 \in (1 - 1/(2k), 1)$  (cf. Definition 4.2.1).

**Example 4.9.9 (Generalized Hermite kernels)** Other examples of generalized Hermite kernels and processes can be constructed by considering the following two useful classes of functions. Let  $S_+^k = \{u \in \mathbb{R}_+^k : \|u\|_2 = 1\}$ , where  $\|u\|_2$  denotes the Euclidean norm. We say that a generalized Hermite kernel  $g$  is of *Class (B)* (B stands for “boundedness”), if on  $S_+^k$ , it is continuous a.e. and bounded. Consequently,

$$|g(u)| \leq \|u\|_2^\alpha g(u/\|u\|_2) \leq c\|u\|_2^\alpha,$$

for some  $c > 0$ . We say that a generalized Hermite kernel  $g$  on  $\mathbb{R}_+^k$  having homogeneity exponent  $\alpha$  is of *Class (L)* (L stands for “limit” as in “limit theorems”), if

1.  $g$  is continuous a.e. on  $\mathbb{R}_+^k$ ;
2.  $|g(u)| \leq g^*(u)$  a.e.  $u \in \mathbb{R}_+^k$ , where  $g^*$  is a finite linear combination of non-symmetric Hermite kernels:  $\prod_{j=1}^k u_j^{\gamma_j}$ , where  $\gamma_j \in (-1, -1/2)$ ,  $j = 1, \dots, k$ , and  $\sum_{j=1}^k \gamma_j = \alpha \in (-k/2 - 1/2, -k/2)$ .

For example,  $g^*(u)$  could be  $u_1^{-3/4} u_2^{-5/8} + u_1^{-9/16} u_2^{-13/16}$  if  $k = 2$ . In this case,  $\alpha = -11/8$ .

Note that Class (B) does not include the original Hermite kernel (4.9.12) but also that Class (L) does. In fact, one can show that Class (L) contains Class (B) (Exercise 4.10 and Proposition 3.20 in Bai and Taqqu [77]). It can also be checked that if two functions  $g_1$  and  $g_2$  on  $\mathbb{R}_+^k$  satisfy Condition 2 of Class (L) above, then  $\int_{\mathbb{R}_+^k} |g_1(u)g_2(1+u)|du < \infty$  automatically holds (Exercise 4.10). In particular, the functions of Class (L) (and hence Class (B)) automatically satisfy the condition (4.9.4) of the generalized Hermite kernels.

Suppose  $g(u) = \|u\|^\alpha$ , where  $\alpha \in (-1/2 - k/2, -k/2)$ . This  $g$  belongs to Class (B) and thus also Class (L). Another example of Class (B):

$$g(u) = \frac{\prod_{j=1}^k u_j^{a_j}}{\sum_{j=1}^k u_j^b},$$

where  $a_j > 0$  and  $b > 0$ , yielding a homogeneity exponent  $\alpha = \sum_{j=1}^k a_j - b \in (-1/2 - k/2, -k/2)$ . Another example of Class (L) but not (B):

$$g(u) = g_0(u) \vee \left( \prod_{j=1}^k u_j^{\alpha/k} \right).$$

where  $g_0(u) > 0$  is any generalized Hermite kernel of Class (B) on  $\mathbb{R}_+^k$  with homogeneity exponent  $\alpha$ .

We conclude this section by addressing the following interesting question. According to Corollary 4.9.4, the generalized Hermite process introduced above admits a self-similarity parameter  $H > 1/2$  only. Are there  $H$ -self-similar processes with stationary increments in a higher-order Wiener chaos having  $H < 1/2$ ? To obtain such a process with  $0 < H < 1/2$ , we consider the following fractionally filtered kernel:

$$h_t^\beta(u) = \int_{\mathbb{R}} l_t^\beta(s) g(s-u) 1_{\{s>u\}} ds, \quad t \in \mathbb{R}, \quad (4.9.13)$$

where  $g$  is a generalized Hermite kernel defined in Definition 4.9.2 with homogeneity exponent  $\alpha \in (-k/2 - 1/2, -k/2)$ , and

$$l_t^\beta(s) = \frac{1}{\beta} \left( (t-s)_+^\beta - (-s)_+^\beta \right), \quad \beta \neq 0. \quad (4.9.14)$$

One can extend it to  $\beta = 0$  by writing  $l_t^0(s) = 1_{[0,t)}(s)$ , but this would lead us back to the generalized Hermite process case. We hence assume throughout that  $\beta \neq 0$ . The following proposition gives the range of  $\beta$  for which  $I_k(h_t^\beta)$  is well-defined.

**Proposition 4.9.10** *If*

$$-1 < -\alpha - \frac{k}{2} - 1 < \beta < -\alpha - \frac{k}{2} < \frac{1}{2}, \quad \beta \neq 0, \quad (4.9.15)$$

*then*  $h_t^\beta \in L^2(\mathbb{R}^k)$ .

*Proof* Observe that, by making the changes of variables  $s = s_1$ ,  $v = s_2 - s_1$  and  $w = s_1 - u$ , and then  $w = vy$  below,

$$\begin{aligned} \int_{\mathbb{R}^k} h_t^\beta(u)^2 du &\leq 2 \int_{-\infty}^{\infty} ds_1 \int_{s_1}^{\infty} ds_2 \int_{\mathbb{R}^k} du l_t(s_1) l_t(s_2) |g(s_1 - u) g(s_2 - u)| 1_{\{s_1 > u\}} \\ &= 2 \int_{-\infty}^{\infty} ds \int_0^{\infty} dv \int_{\mathbb{R}_+^k} dw l_t(s) l_t(s+v) |g(w) g(v+w)| \\ &= 2 \int_{-\infty}^{\infty} ds l_t^\beta(s) \int_0^{\infty} l_t^\beta(s+v) v^{2\alpha+k} dv \int_{\mathbb{R}_+^k} dy |g(y) g(1+y)|. \end{aligned}$$

We thus focus on showing  $\int_{-\infty}^{\infty} ds l_t^\beta(s) \int_0^{\infty} l_t^\beta(s+u) u^{2\alpha+k} du < \infty$ . For any  $c > 0$ , we have

$$\int_0^c (c-s)^{\gamma_1} s^{\gamma_2} ds = c^{\gamma_1+\gamma_2+1} \int_0^1 (1-s)^{\gamma_1} s^{\gamma_2} ds = c^{\gamma_1+\gamma_2+1} B(\gamma_1+1, \gamma_2+1),$$

for all  $\gamma_1, \gamma_2 > -1$ . So by noting that  $\beta > -1$  and  $2\alpha + k > -1$ , we have

$$\begin{aligned} \int_0^{\infty} l_t^\beta(s+u) u^{2\alpha+k} du &= \frac{1}{\beta} \int_0^{\infty} \left( (t-s-u)_+^\beta - (-s-u)_+^\beta \right) u^{2\alpha+k} du \\ &= \frac{1}{\beta} \left( \int_0^{t-s} (t-s-u)^\beta u^{2\alpha+k} du + \int_0^{-s} (-s-u)^\beta u^{2\alpha+k} du \right) \\ &= \frac{B(\beta+1, 2\alpha+k+1)}{\beta} \left( (t-s)_+^{\beta+\delta} - (-s)_+^{\beta+\delta} \right), \end{aligned}$$

where

$$\delta = 2\alpha + k + 1 \in (0, 1). \quad (4.9.16)$$

We thus want to determine when the following holds:

$$\int_{\mathbb{R}} \left( (t-s)_+^\beta - (-s)_+^\beta \right) \left( (t-s)_+^{\beta+\delta} - (-s)_+^{\beta+\delta} \right) ds < \infty. \quad (4.9.17)$$

Suppose  $t > 0$ . The potential integrability problems appear near  $s = -\infty, 0, t$ . Near  $s = -\infty$ , the integrand behaves like  $|s|^{2\beta+\delta-2}$ , and thus we need  $2\beta + \delta - 2 < -1$ ; near  $s = 0$ , the integrand behaves like  $|s|^{2\beta+\delta}$ , and thus  $2\beta + \delta > -1$ ; near  $s = t$ , the integrand behaves like  $|t-s|^{2\beta+\delta}$ , and thus again  $2\beta + \delta > -1$ . In view of (4.9.16), these requirements are satisfied by (4.9.15).  $\square$

The next result is an immediate corollary of Propositions 4.9.1 and 4.9.10.

**Corollary 4.9.11** *The process defined by  $Z^\beta(t) := I_k(h_t^\beta)$ ,  $t \in \mathbb{R}$ , with  $h_t^\beta$  given in (4.9.13), namely,*

$$\begin{aligned} Z^\beta(t) &= \int_{\mathbb{R}^k}' \int_0^t \frac{1}{\beta} \left( (t-s)_+^\beta - (-s)_+^\beta \right) g(s-u_1, \dots, s-u_k) 1_{\{s>u_1, \dots, s>u_k\}} ds B(du_1) \dots B(du_k), \end{aligned} \quad (4.9.18)$$

*is an  $H$ -self-similar process having stationary increments with*

$$H = \alpha + \beta + \frac{k}{2} + 1 \in (0, 1). \quad (4.9.19)$$

**Remark 4.9.12** To get the antipersistent case  $H < 1/2$ , choose

$$\beta \in \left( -\alpha - \frac{k}{2} - 1, -\alpha - \frac{k}{2} - \frac{1}{2} \right).$$

## 4.10 Exercises

The symbols\* and\*\* next to some exercises are explained in Section 2.12.

**Exercise 4.1** Prove the relation (4.1.7) for Hermite polynomials.

**Exercise 4.2** For symmetric  $f \in L^2(E^2, m^2)$ , use the diagram formulae in Propositions 4.3.4 and 4.3.20 to obtain an expression for  $\mathbb{E}I_2(f)^3$ . Argue, in particular, that  $\mathbb{E}I_2(f)^3$  is nonzero and hence  $I_2(f)$  is non-symmetric in general.

**Exercise 4.3** If symmetric  $f_i \in L^2(E, m)$ ,  $i = 1, \dots, 4$ , use the diagram formulae in Propositions 4.3.4 and 4.3.20 to obtain an expression for  $\mathbb{E}I_1(f_1)I_1(f_2)I_1(f_3)I_1(f_4)$ .

**Exercise 4.4** Show that the relation (4.4.11) is equivalent to the relation (4.4.2). The former relation appears in Taqqu [944].

**Exercise 4.5** By following the hint given in the text, prove the relation (4.5.10).

**Exercise 4.6** Prove the two relations in (4.7.6).

**Exercise 4.7\*** Use the Markov inequality to prove Corollary 4.7.7, (ii). *Hint:* See Corollary 4.4 in Veillette and Taqu [979]. (The parts (i) and (iii) of Corollary 4.7.7 can be proved using auxiliary results on infinitely divisible distributions – see Corollaries 4.3 and 4.5 in Veillette and Taqu [979].)

**Exercise 4.8\*\*** Prove the conjecture stated in Section 4.7 that the CDFs of the Rosenblatt distribution  $Z_D$  cross at the same point for all  $D$ s, as suggested by Figure 4.13.

**Exercise 4.9** Prove Proposition 4.9.1.

**Exercise 4.10** Prove the following properties of Classes (B) and (L) introduced in Example 4.9.9. (i) Show that Class (L) contains Class (B). (ii) If two functions  $g_1$  and  $g_2$  on  $\mathbb{R}_+^k$  satisfy Condition 2 of Class (L), then  $\int_{\mathbb{R}_+^k} |g_1(u)g_2(1+u)|du < \infty$  automatically holds.

**Exercise 4.11\*** If  $X$  has a normal distribution, show that  $X^3$  is indeterminate; that is, its distribution is not determined by the moments. *Hint:* Suppose without loss of generality that  $X$  has the density  $e^{-x^2}/\sqrt{\pi}$  and let  $d(x)$  be the density of  $X^3$ . Show then that the densities  $d(x)$  and  $d(x)\{1 + r(\cos(\sqrt{3}|x|^{2/3}) - \sqrt{3}\sin(\sqrt{3}|x|^{2/3}))\}$  with  $|r| \leq 1/2$  have all their moments equal. See Proposition 1 in Berg [129]. (The fact that higher-order Hermite polynomials of normal variables or multiple integrals with respect to Gaussian measures may not be determinate is well known. See, e.g., Slud [910].)

## 4.11 Bibliographical Notes

**Section 4.1:** Hermite polynomials are special orthogonal polynomials that play important roles in Statistics and Probability, but also in many other areas. The monographs by Andrews, Askey, and Roy [35] and Lebedev [601] are excellent sources on the properties of Hermite and related polynomials. There are many related definitions of Hermite polynomials as noted in a footnote in Section 4.1. It is important therefore to always check which definition is used.

The connections between Hermite polynomials, multiple stochastic integrals and Gaussian variables were recognized from the very start, e.g., by Itô [524], and have been exploited and documented widely since, e.g., in Nualart [769], Major [672, 673]. The latter reference, in particular, contains interesting historical accounts.

**Section 4.2:** The term “Hermite processes” was coined by Murad Taqu. The integral representations of Hermite processes in the time and spectral domains appeared first in Dobrushin and Major [321], Taqu [945]. The integral representation on an interval was derived following Pipiras and Taqu [819], though the representation goes back at least to Tudor [962] in the case of the Rosenblatt process, and to Nourdin, Nualart, and Tudor [766] in the general case of Hermite processes. It was subsequently used in, for example, Breton and Nourdin

[183], Chronopoulou, Tudor, and Viens [240, 241]. Representations of fractional Brownian motion are discussed further in the notes to Section 6.2 below.

**Section 4.3:** The presentation follows parts of Peccati and Taqqu [797] whose approach originated with Rota and Wallstrom [865]. See also the books of Janson [529], Major [673], with the latter containing interesting historical notes. In the context of Hermite processes, the diagram and related formulae have originated with Dobrushin [320], Taqqu [944], though their variants have also been known well earlier, especially in the literature of Mathematical Physics in connection to Feynman's diagrams (e.g., Polyak [828]).

**Section 4.4:** The moments of Hermite processes were studied originally by Taqqu [944].

**Section 4.5:** The representation of multiple integrals of order two in Lemma 4.5.2 goes back at least to Varberg [974]. Multiple integrals of order greater than two are known to be quite different from those of order two, for example, their cumulants do not typically determine the distribution anymore (Slud [910]).

**Section 4.6:** The Rosenblatt process was introduced by Taqqu [943], naming it after Rosenblatt [860] who considered its marginal distribution. The history and the developments behind the Rosenblatt process are described in Taqqu [946].

**Section 4.7:** As noted in the section, its material is drawn from the work of Veillette and Taqqu [979, 980]. The Rosenblatt distribution was also studied by Albin [18], Maejima and Tudor [669].

**Section 4.9:** The material of this section is based on Bai and Taqqu [77] who developed the notion of “generalized Hermite processes.” These processes were introduced by Maejima and Tudor [668]. Similar types of self-similar processes with stationary increments of higher-order Wiener chaos were also studied by Mori and Oodaira [732], Arras [44].